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HÖRMANDER TYPE FOURIER MULTIPLIER THEOREM AND NIKOLSKII INEQUALITY ON QUANTUM TORI, AND APPLICATIONS

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Abstract. In this paper, we study Hörmander type Fourier multiplier theorem and the Nikolskii inequality on quantum tori. On the way to obtain these results, we also prove some classical inequalities such as Paley type, Hausdorff-Young-Paley, Hardy-Littlewood, and Logarithmic Sobolev inequalities on quantum tori. As applications we establish embedding theorems between Sobolev, Besov spaces as well as embeddings between Besov and Wiener and Beurling spaces on quantum tori. We also analyse β-versions of Wiener and Beurling spaces and their embeddings, and interpolation properties of all these spaces on quantum tori. As an applications of the analysis, we also derive a version of the Nash inequality, and the time decay for solutions of a heat type equation.

Contents

1. Introduction 2
2. Preliminaries 3
2.1. Quantum tori 3
2.2. Fourier multipliers on quantum tori 6
3. Hausdorff-Young inequality and its inverse form 7
4. Hausdorff–Young–Paley Inequality 8
5. Hörmander multiplier theorem on quantum tori 14
6. Sobolev and logarithmic Sobolev inequalities on quantum tori 17
6.1. Nash type inequality on quantum tori 20
7. Nikolskii inequality 23
8. Besov spaces on quantum tori 27
9. Wiener and β-Wiener spaces on quantum tori 31
10. Beurling and β-Beurling spaces on quantum tori 34
11. Interpolation 36
11.1. Conflict of interest 40
11.2. Data availability 40
12. Acknowledgements 40
References 40

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1. Introduction

Quantum tori $\mathbb{T}^d_\theta$ is one of the main objectives in Alain Connes' non-commutative geometry (see [12] and [13], [19, Chapter 12]) and in non-commutative harmonic analysis [21, 46, 47, 50, 54]. This is also known as a non-commutative torus and irrational rotation algebra. In the last decades, there has been substantial body of work on generalising the methods of classical harmonic analysis on ordinary torus to quantum tori (or non-commutative torus). In particular, on a non-commutative torus (defined in terms of an arbitrary antisymmetric real $d\times d$ matrix θ), it is possible to define analogues of many of the tools of harmonic analysis, such as differential operators and function spaces [34, 54]. In this setting, many of classical inequalities in harmonic analysis and function spaces are seen in a new light.

The study of Fourier multipliers has been attracting attention of many researchers for more than a century. This is related to numerous applications in harmonic analysis, in particular, in PDEs. One of the important questions in this field is to understand the $L^p\to L^q$ boundedness of Fourier multipliers. In the case $p=q$, remarkable results were obtained by Marcinkiewicz [26], [33] and Mikhlin [35], which were improved later by Stein [51] and Hörmander [25].

For the case $L^p\to L^q$, there is another classical result of Hörmander available in [25, Theorem 1.11]. If $\mathcal{S}(\mathbb{R}^d)$ is the Schwartz class and T_g is the Fourier multiplier with the symbol $g\in\mathcal{S}(\mathbb{R}^d)$, which is given by the multiplication on the Fourier transform side

$$\widehat{T_g(f)}=g\cdot\widehat{f},\quad f\in\mathcal{S}(\mathbb{R}^d),$$

where $\widehat{f}$ is the Fourier transform

$$\widehat{f}(t)=(2\pi)^{-d/2}\int_{\mathbb{R}^d}f(s)e^{-i(t,s)}ds,\quad t\in\mathbb{R}^d,$$

then Hörmander's result asserts as follows:

Theorem 1.1. [25, Theorem 1.11] *If $1<p\leq 2\leq q<\infty$, then the Fourier multiplier T_g with the symbol $g:\mathbb{R}^d\to\mathbb{C}$ extends to a bounded map from $L^p(\mathbb{R}^d)$ to $L^q(\mathbb{R}^d)$ and we have*

$$\|T_g\|_{L^p(\mathbb{R}^d)\to L^q(\mathbb{R}^d)}\lesssim\sup_{\lambda>0}\lambda\left(\int\limits_{\substack{x\in\mathbb{R}^d\\ |g(x)|\geq\lambda}}dx\right)^{\frac{1}{p}-\frac{1}{q}}. \tag{1.1}$$

It can be found from [18, p. 303] that under the same conditions on p,q, and r, for a sequence of complex numbers $g=\{g(k)\}_{k\in\mathbb{Z}^d}$ and the corresponding Fourier series multiplier T_g, the same result as in Theorem 1.1 holds. There are also other works in this direction, see e.g. [1, 4, 10, 29, 30, 31, 38, 40, 41, 48, 57]. For example, in [3], authors obtained L^p-L^q Hörmander type theorem in the context of locally compact groups with applications such as the L^p-L^q boundedness of the heat semigroup and Sobolev embedding theorem in which our results are motivated. Fourier multipliers on quantum tori and their $L^p\to L^p$ estimates, as well as the completely boundedness properties were studied deeply in [11, 46, 54] (see also [52] in more general settings of quasi-Banach symmetric spaces).

In this work, we study the $L^p\to L^q$ boundedness of the Fourier multipliers on quantum tori. In other words, we obtain a quantum analogue of the above Theorem 1.1 on quantum tori (see Theorem 5.1). On the way to achieve our goal, we prove a number of

classical inequalities such as Paley, Hausdorff-Young-Paley, Hardy-Littlewood, and Logarithmic Sobolev inequalities on quantum tori which also have further applications in PDEs. As a simple application of the Hörmander type theorem we obtain an $L^p \to L^q$ estimate of the heat semi-group (see Corollary 5.2) and Sobolev type inequality (see Theorem 6.2) on quantum tori. As an application of our Nash-type inequality we compute the decay rate for the heat equation for the sub-Laplacian in our setting as in the classical case.

As a second main topic we analyse the families of Besov, Wiener and Beurling spaces on quantum tori. To a large extent, the analysis is based on establishing an appropriate version of Nikolskii's inequality, which is also known as the reverse Hölder inequality, in this setting. The classical Nikolskii inequality for trigonometric polynomials on the circle was given by [36] (see also [49, Chapter III.3.2, p. 147]). This inequality plays an important role in investigating basic properties of different function spaces [37, 49], in approximation theory [16], and in obtaining embedding theorems [17, 49]. The Nikolskii inequality for the spherical polynomials was proved in [27, Theorem 1], see also [15]. Furthermore, in [43] and [44] Pesenson obtained a very general version of Bernstein-Nikolskii type inequalities on compact homogeneous and non-compact symmetric spaces, respectively. In this work, we prove the Nikolskii inequality for trigonometric polynomials in non-commutative L^p- spaces associated with a quantum tori. In the special case when $\theta = 0$, our result Theorem 7.2 coincides with [49, Proposition, p. 147] and extends Pesenson's result [43] to a wider range of indices $0 < p \leq q \leq \infty$ with a completely different approach of proof. Our method is based on the idea in [39], where the authors studied similar problems on compact homogeneous manifolds. As applications of the Nikolskii inequality we establish embedding theorems between Besov spaces, as well as embeddings between Besov and Wiener and Beurling spaces on quantum tori. Generally, Besov spaces on quantum tori and their embedding properties were studied recently in [54, Chapter V and VI] in the case when $1 \leq p, q \leq \infty$. We study similar questions extending the range of indices to $0 < p, q \leq \infty$ by using Nikolskii inequality. Apart from embeddings between Besov spaces that can be obtained by trivial arguments, the Nikolskii inequality allows us to derive a rather complete list of embeddings with respect to all three indices of the space which we address in Theorem 8.3. Finally, we analyse β-versions of Wiener and Beurling spaces and their embeddings, and interpolation properties of all these spaces on quantum tori in Section 11.

2. Preliminaries

2.1. **Quantum tori.** Let $d \geq 2$ be an integer and let θ be a real anti-symmetric (or skew-symmetric) $d \times d$-matrix. We denote by $C(\mathbb{T}^d_\theta)$ a $*$-algebra generated by elements $(U_k)_{1\leq k\leq d}$ satisfying the condition

$$U_k U_m = e^{i\theta_{km}} U_m U_k, \quad U_k U_k^* = U_k^* U_k = 1.$$

The natural Hamel basis in $C(\mathbb{T}^d_\theta)$ is $\{e^\theta_n\}_{n\in\mathbb{Z}^d}$,

$$e^\theta_n = U_1^{n_1} U_2^{n_2} \dots U_d^{n_d}, \quad n = (n_1, \dots, n_d) \in \mathbb{Z}^d. \tag{2.1}$$

Note that

$$e^\theta_n e^\theta_m = e^{-i(\sum\limits_{j<k} \theta_{jk} n_j m_k)} e^\theta_{n+m}, \quad (e^\theta_n)^* = e^{-i(\sum\limits_{j<k} \theta_{jk} n_j n_k)} e^\theta_{-n}. \tag{2.2}$$

Consider a linear functional τ_θ on $C(\mathbb{T}^d_\theta)$ defined by the formula

$$\tau_\theta(e^\theta_n) = \begin{cases} 1 & \text{for} \quad n = 0, \\ 0 & \text{for} \quad n \neq 0. \end{cases} \tag{2.3}$$

By formulas (2.2)-(2.3) and by a simple computation we find some relations which will be useful for our further investigations. Indeed, we have that

$$|\tau_\theta(e^\theta_n e^\theta_m)| \overset{(2.2)}{=} |e^{-i(\sum\limits_{j<k}\theta_{jk}n_jm_k)}\tau_\theta(e^\theta_{n+m})| \overset{(2.3)}{=} |e^{i(\sum\limits_{j<k}\theta_{jk}n_jm_k)}||\tau_\theta(e^\theta_{n+m})| \overset{(2.3)}{=} 1 \tag{2.4}$$

whenever $n+m=0$, and

$$\tau_\theta((e^\theta_n)^* e^\theta_n) \overset{(2.2)}{=} \tau_\theta(e^{-i(\sum\limits_{j<k}\theta_{jk}n_jn_k)} e^\theta_{-n}e^\theta_n) \overset{(2.2)}{=} \tau_\theta(e^\theta_0) \overset{(2.3)}{=} 1. \tag{2.5}$$

Also, $\tau_\theta((e^\theta_n)^* e^\theta_m) = 0$ for $n \neq m$. A polynomial in $C(\mathbb{T}^d_\theta)$ is a finite sum

$$x = \sum_{m\in\mathbb{Z}^d} \alpha(m) e^\theta_m, \quad \alpha(m) \in \mathbb{C}, \quad m \in \mathbb{Z}^d,$$

that is, $\alpha(m) = 0$ for all but finite indices $m \in \mathbb{Z}^d$. The involution algebra of all such polynomials is dense in $C(\mathbb{T}^d_\theta)$. For any polynomial x as above, we define $\tau_\theta(x) = \alpha(0)$, where $0 = (0, \dots, 0)$. Then, τ_θ extends to a faithful tracial state on $C(\mathbb{T}^d_\theta)$. Moreover, we have

$$\begin{aligned} \tau_\theta((\sum_{m\in\mathbb{Z}^d}\alpha(m)e^\theta_m)(\sum_{n\in\mathbb{Z}^d}\beta(n)e^\theta_n)) &= \sum_{m,n\in\mathbb{Z}^d}\alpha(m)\beta(n)\tau_\theta\left(e^\theta_m e^\theta_n\right) \\ &\overset{(2.4)}{=} \sum_{n\in\mathbb{Z}^d} e^{i(\sum\limits_{j<k}\theta_{jk}n_jn_k)}\alpha(-n)\beta(n), \end{aligned}$$

which implies that

$$\tau_\theta(xy) = \tau_\theta(yx), \quad x, y \in C(\mathbb{T}^d_\theta).$$

Let us equip $C(\mathbb{T}^d_\theta)$ with an inner product defined by the formula

$$(x, y) = \tau_\theta(xy^*).$$

Then $(C(\mathbb{T}^d_\theta), (\cdot,\cdot))$ becomes a pre-Hilbert space. Let $L^2(\mathbb{T}^d_\theta)$ be the completion of $(C(\mathbb{T}^d_\theta), (\cdot,\cdot))$ defined by the (nondegenerate) above inner product, which makes $H := L^2(\mathbb{T}^d_\theta)$ a separable Hilbert space. Obviously, $\{e^\theta_n\}_{n\in\mathbb{Z}^d}$ is an orthonormal basis in $L^2(\mathbb{T}^d_\theta)$. We frequently refer to $\{e^\theta_n\}_{n\in\mathbb{Z}^d}$ as a trigonometric basis (with which it coincides for $\theta = 0$). We represent $C(\mathbb{T}^d_\theta)$ on $L^2(\mathbb{T}^d_\theta)$ by left (resp. right) multiplication $\lambda_l(x)y := xy$, and denote by $L^\infty(\mathbb{T}^d_\theta)$ the corresponding weak* closure of $\lambda_l(C(\mathbb{T}^d_\theta))$, a von Neumann algebra with operator norm denoted by $\|\cdot\|_\infty$. Recall that the von Neumann algebra $L^\infty(\mathbb{T}^d_\theta)$ is hyper finite [54]. For

$x = \sum_{n\in\mathbb{Z}^d} \alpha(n)e_n^\theta$, we have

$$\begin{aligned}\tau_\theta(x^*x) &= \tau_\theta\left(\left(\sum_{m\in\mathbb{Z}^d}\alpha(m)e_m^\theta\right)^*\left(\sum_{n\in\mathbb{Z}^d}\alpha(n)e_n^\theta\right)\right)\\ &= \sum_{m,n\in\mathbb{Z}^d}\overline{\alpha(m)}\alpha(n)\tau_\theta\left((e_m^\theta)^*e_n^\theta\right)\\ &= \sum_{n\in\mathbb{Z}^d}|\alpha(n)|^2. \qquad (2.6)\end{aligned}$$

Therefore, $\tau_\theta(x^*x) = 0$ implies that $x = 0$. Both τ_θ and λ_l extend continuously to $L^\infty(\mathbb{T}_\theta^d)$, giving a faithful tracial state $\tau_\theta : L^\infty(\mathbb{T}_\theta^d) \to \mathbb{C}$ and a faithful representation (injective $*$-homomorphism) $\lambda_l : L^\infty(\mathbb{T}_\theta^d) \to \mathbb{B}(L^2(\mathbb{T}_\theta^d))$, where $\mathbb{B}(L^2(\mathbb{T}_\theta^d))$ is the algebra of all linear bounded operators on $L^2(\mathbb{T}_\theta^d)$. We identify $L^\infty(\mathbb{T}_\theta^d) \subseteq L^2(\mathbb{T}_\theta^d)$. In fact, the unitaries $U_1, ..., U_d$ can be concretely realized as unitary operators on $L^2(\mathbb{T}^d)$. Indeed, let θ be a real anti-symmetric $d \times d$-matrix and $\theta_1, ..., \theta_d$ be its column vectors. Then, for $j = 1, ..., d$ we define the unitary operators

$$(U_j\xi)(s) = e^{is_j}\xi(s + \pi\theta_j), \quad \xi \in L^2(\mathbb{T}^d),$$

which satisfy the relations,

$$U_kU_j = e^{2\pi i\theta_{jk}}U_jU_k, \quad j, k = 1, ..., d.$$

In this case, our C^*-algebra $C(\mathbb{T}_\theta^d)$ is generated by the above unitaries. For more details, we refer the reader to [21]. If $\theta = 0$, then our spaces $C(\mathbb{T}_\theta^d)$ and $L^\infty(\mathbb{T}_\theta^d)$ coincide with the spaces of continuous functions $C(\mathbb{T}^d)$ and essentially bounded functions $L^\infty(\mathbb{T}^d)$ on the ordinary d-torus $\mathbb{T}^d$. Since $L^\infty(\mathbb{T}_\theta^d)$ is a finite von Neumann algebra, further we always assume that $\tau_\theta(\mathbf{1}) = 1$. The pair $(L^\infty(\mathbb{T}_\theta^d), \tau_\theta)$ is called a non-commutative measure space. As usual, we define $L^p(\mathbb{T}_\theta^d)$ as a non-commutative L^p-space associated with the von Neumann algebra $L^\infty(\mathbb{T}_\theta^d)$ and the trace τ_θ. For more details on non-commutative L^p spaces associated with a general semifinite von Neumann algebra we refer the reader to [32], [45]. Indeed, for any $0 < p < \infty$, we can define the L^p-norm on this space by the Borel functional calculus and the following formula:

$$\|x\|_p = \left(\tau_\theta(|x|^p)\right)^{1/p}, \quad x \in L^\infty(\mathbb{T}_\theta^d),$$

where $|x| := (x^*x)^{1/2}$. The completion of $\{x \in L^\infty(\mathbb{T}_\theta^d) : \|x\|_p < \infty\}$ with respect to $\|\cdot\|_p$ is denoted by $L^p(\mathbb{T}_\theta^d)$. The elements of $L^p(\mathbb{T}_\theta^d)$ are τ_θ-measurable operators like in the commutative case. These are linear densely defined closed (possibly unbounded) affiliated with $L^\infty(\mathbb{T}_\theta^d)$ operators. For $x \in L^1(\mathbb{T}_\theta^d)$ we can define the formal Fourier series of x by

$$x \sim \sum_{n\in\mathbb{Z}^d}\widehat{x}(n)e_n^\theta$$

with $\widehat{x}(n) = \tau_\theta(x(e_n^\theta)^*)$, $n \in \mathbb{Z}^d$. Every element $x \in L^2(\mathbb{T}_\theta^d)$ admits a unique representation of the form

$$x = \sum_{n\in\mathbb{Z}^d}\widehat{x}(n)e_n^\theta, \quad \{\widehat{x}(n)\}_{n\in\mathbb{Z}^d} \in \ell^2(\mathbb{Z}^d), \qquad (2.7)$$

where

$$\widehat{x}(n)=\tau_\theta(x(e_n^\theta)^*),\quad n\in\mathbb{Z}^d. \tag{2.8}$$

Moreover, we have the Plancherel (or Parseval) identity (see, [34, Formula 2.3])

$$\|x\|_{L^2(\mathbb{T}^d_\theta)}=\|\widehat{x}\|_{\ell^2(\mathbb{Z}^d)}. \tag{2.9}$$

Remark 2.1. It is well known from the general theory of linear bounded operators that the composition of two unitary operators and any integer power of unitary operators are also unitary operators. Since e_n^θ is defined by a composition of unitary operators (see (2.1)), it is itself a unitary operator for all $n\in\mathbb{Z}^d$ and we have

$$\|e_n^\theta\|_{L^\infty(\mathbb{T}^d_\theta)}=\|(e_n^\theta)^*\|_{L^\infty(\mathbb{T}^d_\theta)}=1. \tag{2.10}$$

The space $C^\infty(\mathbb{T}^d_\theta)$ is defined to be the set of $x=\sum\limits_{n\in\mathbb{Z}^d}\widehat{x}(n)e_n^\theta$ such that the sequence of Fourier coefficients $\{\widehat{x}(n)\}_{n\in\mathbb{Z}^d}$ has a rapid decay (the sequence $\{\widehat{x}(n)\}_{n\in\mathbb{Z}^d}$ is eventually dominated by the reciprocal of any polynomial, that is for any $m\in\mathbb{Z}_+$ there is $c_m>0$ such that $|\widehat{x}(n)|\le c_m(1+|n|)^{-m}, n\in\mathbb{Z}^d$, where $|\cdot|$ is Euclidean length on $\mathbb{Z}^d$). This is a weak*-dense $*$-subalgebra of $C(\mathbb{T}^d_\theta)$ and contains all polynomials. We may consider $C^\infty(\mathbb{T}^d_\theta)$ as a non-commutative deformation of the space of smooth functions on $\mathbb{T}^d$.

Throughout this paper, we write $\mathcal{A}\lesssim\mathcal{B}$ if there is a constant $c>0$ depending only on parameters of spaces such that $\mathcal{A}\le c\mathcal{B}$. We write $\mathcal{A}\asymp\mathcal{B}$ if both $\mathcal{A}\lesssim\mathcal{B}$ and $\mathcal{A}\gtrsim\mathcal{B}$ hold, possibly with different constants.

2.2. **Fourier multipliers on quantum tori.** Let $g:\mathbb{Z}^d\to\mathbb{C}$ be a bounded scalar function on $\mathbb{Z}^d$. The Fourier multiplier T_g is defined on x with the symbol g by

$$T_g(x)=\sum_{n\in\mathbb{Z}^d}g(n)\widehat{x}(n)e_n^\theta,\quad x\in L^2(\mathbb{T}^d_\theta). \tag{2.11}$$

By virtue of the Plancherel identity (2.9) and the uniqueness of the representation (2.7), T_g indeed defines a bounded linear operator on $L^2(\mathbb{T}^d_\theta)$ and the above series converges in $L^2(\mathbb{T}^d_\theta)$-sense. If g is unbounded, we may define T_g on the dense subspace of $L^2(\mathbb{T}^d_\theta)$ of those x with finitely many non-zero Fourier coefficients.

Definition 2.2. Let $1\le p,q\le\infty$. For a bounded linear operator $T:L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)$ we denote by $T^*:L^{q'}(\mathbb{T}^d_\theta)\to L^{p'}(\mathbb{T}^d_\theta)$ its adjoint operator defined by

$$(T(x),y):=\tau_\theta(T(x)y^*)=\tau_\theta(x(T^*(y))^*)=(x,T^*(y)) \tag{2.12}$$

for all $x\in L^p(\mathbb{T}^d_\theta)$ and $y\in L^{q'}(\mathbb{T}^d_\theta)$ (or in a dense subspace of it).

Lemma 2.3. *Let $1<p,q<\infty$ and let $T_g:L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)$ be the Fourier multiplier defined by (2.11) with the symbol g. Then its adjoint $T_g^*:L^{q'}(\mathbb{T}^d_\theta)\to L^{p'}(\mathbb{T}^d_\theta)$ equals to $T_{\overline{g}}$, where $\overline{g}$ is the complex conjugate of the function g, in the sense of Definition 2.2.*

Proof. By formulas (2.3) and (2.11) we write

$$
\begin{aligned}
(x, T_g^*(y)) &\overset{(2.12)}{=} \tau_\theta(x(T_g^*(y))^*) = \tau_\theta(T_g(x)y^*) \\
&\overset{(2.11)}{=} \tau_\theta\Big(\sum_{n\in\mathbb{Z}^d} g(n)\widehat{x}(n)e_n^\theta\big(\sum_{m\in\mathbb{Z}^d}\widehat{y}(m)e_m^\theta\big)^*\Big) \\
&\overset{(2.3)}{=} \sum_{m,n\in\mathbb{Z}^d} g(n)\widehat{x}(n)\overline{\widehat{y}(m)}\tau_\theta\big(e_n^\theta(e_m^\theta)^*\big) \\
&= \tau_\theta\Big(\sum_{n\in\mathbb{Z}^d}\widehat{x}(n)e_n^\theta\sum_{m\in\mathbb{Z}^d} g(m)\overline{\widehat{y}(m)}(e_m^\theta)^*\Big) \\
&= \tau_\theta\Big(\sum_{n\in\mathbb{Z}^d}\widehat{x}(n)e_n^\theta\big(\sum_{m\in\mathbb{Z}^d}\overline{g(m)}\widehat{y}(m)e_m^\theta\big)^*\Big) \\
&= (x, T_{\overline{g}}(y)), \quad x, y \in C^\infty(\mathbb{T}_\theta^d),
\end{aligned}
$$

where

$$
T_{\overline{g}}(y) := \sum_{m\in\mathbb{Z}^d}\overline{g(m)}\widehat{y}(m)e_m^\theta, \quad y \in C^\infty(\mathbb{T}_\theta^d),
$$

and we used that $\tau_\theta\big(e_n^\theta(e_m^\theta)^*\big) = 0$ for $n \neq m$. Since $C^\infty(\mathbb{T}_\theta^d)$ is dense in $L^p(\mathbb{T}_\theta^d)$ (see [54, Proposition 2.7. (ii), p. 21]) for $1 \leq p \leq \infty$, we have $T_g^* = T_{\overline{g}}$. □

3. Hausdorff-Young inequality and its inverse form

In this section, we obtain the Hausdorff-Young inequality and its inverse form for the non-commutative torus.

Theorem 3.1. *(Hausdorff-Young inequality). Let $1 \leq p \leq 2$ and $\frac{1}{p} + \frac{1}{p'} = 1$. Then for any $x \in L^p(\mathbb{T}_\theta^d)$ we have*

$$
\|\widehat{x}\|_{\ell^{p'}(\mathbb{Z}^d)} \leq \|x\|_{L^p(\mathbb{T}_\theta^d)}. \tag{3.1}
$$

Proof. We denote by T the operator defined by $T(x) = \{\widehat{x}(n)\}_{n\in\mathbb{Z}^d}$ for $x \in L^p(\mathbb{T}_\theta^d)$, $1 \leq p \leq 2$. Clearly, T is linear due to the linearity of the trace τ_θ defined by (2.3). Furthermore, by formulas (2.10) and (2.8) and using the non-commutative Hölder inequality (see [53, Theorem 5.2.9]), we obtain

$$
\begin{aligned}
\|T(x)\|_{\ell^\infty(\mathbb{Z}^d)} &= \|\widehat{x}\|_{\ell^\infty(\mathbb{Z}^d)} \overset{(2.8)}{=} \sup_{n\in\mathbb{Z}^d}\left|\tau_\theta\left(x(e_n^\theta)^*\right)\right| \\
&\leq \sup_{n\in\mathbb{Z}^d}\|(e_n^\theta)^*\|_{L^\infty(\mathbb{T}_\theta^d)}\|x\|_{L^1(\mathbb{T}_\theta^d)} \overset{(2.10)}{\leq} \|x\|_{L^1(\mathbb{T}_\theta^d)},
\end{aligned}
$$

for all $n \in \mathbb{Z}^d$. Moreover, by the Plancherel identity (2.9), we have

$$
\|T(x)\|_{\ell^2(\mathbb{Z}^d)} = \|\widehat{x}\|_{\ell^2(\mathbb{Z}^d)} \overset{(2.9)}{=} \|x\|_{L^2(\mathbb{T}_\theta^d)}, \quad x \in L^2(\mathbb{T}_\theta^d).
$$

Thus, T is bounded from $L^1(\mathbb{T}_\theta^d)$ into $\ell^\infty(\mathbb{Z}^d)$ and from $L^2(\mathbb{T}_\theta^d)$ into $\ell^2(\mathbb{Z}^d)$, with the operator norms at most 1. If $\eta = 2/p'$, then $0 \leq \eta \leq 1$ and we have $\frac{1}{p} = \frac{1-\eta}{1} + \frac{\eta}{2}$ and $\frac{1}{p'} = \frac{1-\eta}{\infty} + \frac{\eta}{2}$. Hence, the assertion follows from the Riesz-Thorin Interpolation Theorem (see, [45, Theorem 2.1]). □

The following is an anlogue of the classical Riesz-Fischer Theorem for Fourier Series on the non-commutative torus.

Theorem 3.2. *(Riesz-Fischer Theorem for Fourier Series) Let $\{\xi(n)\}_{n\in\mathbb{Z}^d}$ be any complex valued sequence for which $\sum_{n\in\mathbb{Z}^d}|\xi(n)|^2$ converges. Then there exists an operator $x\in L^2(\mathbb{T}^d_\theta)$ with $\widehat{x}(n)=\xi(n)$ for all $n\in\mathbb{Z}^d$. In particular,*

$$\sum_{n\in\mathbb{Z}^d}|\xi(n)|^2=\|x\|^2_{L^2(\mathbb{T}^d_\theta)}.$$

Proof. The proof is similar to the commutative case, therefore, is omitted (see [24, Theorem 16.39, p. 250]). □

The following result shows that inequality (3.1) remains true (with the reversed inequality) also for $2\leq p\leq\infty$.

Theorem 3.3. *If $2\leq p\leq\infty$ and if $\xi=\{\xi(n)\}_{n\in\mathbb{Z}^d}\in\ell^{p'}(\mathbb{Z}^d)$, then the operator x with $\widehat{x}(n)=\xi(n)$ for all $n\in\mathbb{Z}^d$ belongs to $L^p(\mathbb{T}^d_\theta)$, and we have*

$$\|x\|_{L^p(\mathbb{T}^d_\theta)}\leq\|\widehat{x}\|_{\ell^{p'}(\mathbb{Z}^d)} \tag{3.2}$$

where $\frac{1}{p}+\frac{1}{p'}=1$.

Proof. The idea of the proof is similar to that of Theorem 3.1. But, the main question is to define the operator correctly. Let us define the map T_1 by

$$T_1:\xi=\{\xi(n)\}_{n\in\mathbb{Z}^d}\mapsto x:=\sum_{n\in\mathbb{Z}^d}\xi(n)e^\theta_n$$

which is well defined on $\ell^1(\mathbb{Z}^d)$ into $L^\infty(\mathbb{T}^d_\theta)$. On the other hand, Theorem 3.2 exhibits an isometry T_2 from $\ell^2(\mathbb{Z}^d)$ onto $L^2(\mathbb{T}^d_\theta)$ by $T_2\xi=x$ with the property $\xi(n)=\widehat{x}(n)$ for all $n\in\mathbb{Z}^d$. Therefore, if $\xi\in\ell^2(\mathbb{Z}^d)$, then $T_1\xi$ and $T_2\xi$ have the same Fourier coefficients and hence coincide. Since $\ell^1(\mathbb{Z}^d)$ is dense in $\ell^2(\mathbb{Z}^d)$, the operator T_2 is therefore the unique extension of T_1 to an isometry of $\ell^2(\mathbb{Z}^d)$ into $L^2(\mathbb{T}^d_\theta)$. This extension defines the Fourier coefficients with respect to $\mathbb{Z}^d$. Interpolating as in Theorem 3.1, we obtain that if $\xi=\{\xi(n)\}_{n\in\mathbb{Z}^d}\in\ell^{p'}(\mathbb{Z}^d)$, then the operator x with $\xi(n)=\widehat{x}(n)$ for all $n\in\mathbb{Z}^d$ belongs to $L^p(\mathbb{T}^d_\theta)$, and

$$\|x\|_{L^p(\mathbb{T}^d_\theta)}\leq\|\xi\|_{\ell^{p'}(\mathbb{Z}^d)}.$$

In other words, we obtain (3.2). □

4. Hausdorff–Young–Paley Inequality

Here we give an analogue of the Hausdorff–Young–Paley inequality on $\mathbb{T}^d_\theta$ which is a generalization of the Hausdorff–Young inequality.

Theorem 4.1. *(Paley-type inequality). Let $1<p\leq 2$ and let $h:\mathbb{Z}^d\to\mathbb{R}_+$ be a strictly positive function such that*

$$M_h:=\sup_{\lambda>0}\lambda\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\geq\lambda}}1<\infty. \tag{4.1}$$

Then for any $x \in L^p(\mathbb{T}^d_\theta)$ we have the following estimate

$$\left(\sum_{n\in\mathbb{Z}^d} |\widehat{x}(n)|^p h^{2-p}(n)\right)^{\frac{1}{p}} \leq c_p M_h^{\frac{2-p}{p}} \|x\|_{L^p(\mathbb{T}^d_\theta)}, \tag{4.2}$$

where $c_p > 0$ is a constant independent of x.

Proof. Define a measure μ on $\mathbb{Z}^d$ such that $\mu(n) := h^2(n) > 0$ for $n \in \mathbb{Z}^d$. Then it can be proved that the space $\ell^p(\mathbb{Z}^d, \mu)$ is a Banach space of all complex valued sequences $a = \{a(n)\}_{n\in\mathbb{Z}^d}$ with the norm

$$\|a\|_{\ell^p(\mathbb{Z}^d,\mu)} := \left(\sum_{n\in\mathbb{Z}^d} |a(n)|^p h^2(n)\right)^{\frac{1}{p}} < \infty.$$

Now, define an operator $A : L^p(\mathbb{T}^d_\theta) \to \ell^p(\mathbb{Z}^d, \mu)$ by the formula

$$Ax = \left\{\frac{\widehat{x}(n)}{h(n)}\right\}_{n\in\mathbb{Z}^d}.$$

It is clear that A is a sub-linear (or quasi-linear) operator. Next, we will show that A is well-defined and bounded from $L^p(\mathbb{T}^d_\theta)$ to $\ell^p(\mathbb{Z}^d, \mu)$ for $1 < p \leq 2$. In other words, we claim that we have the estimate (4.2) with the condition (4.1). First, we will show that A is of weak type $(2, 2)$ and of weak-type $(1, 1)$. Recall that the distribution function $d_{Ax}(\lambda)$, $\lambda > 0$ with respect to the weight $h^2(n) > 0$ is defined by

$$d_{Ax}(\lambda) := \mu\{n \in \mathbb{Z}^d : |(Ax)(n)| > \lambda\} = \sum_{\substack{n\in\mathbb{Z}^d \\ |(Ax)(n)|>\lambda}} h^2(n).$$

We show that

$$d_{Ax}(\lambda) \leq c_2 \frac{\|x\|^2_{L^2(\mathbb{T}^d_\theta)}}{\lambda^2} \quad \text{with} \quad c_2 = 1, \tag{4.3}$$

and

$$d_{Ax}(\lambda) \leq \frac{c_1\|x\|_{L^1(\mathbb{T}^d_\theta)}}{\lambda} \quad \text{with} \quad c_1 = 2M_h. \tag{4.4}$$

Indeed, we first prove inequality (4.3). It follows from the Chebyshev inequality and Plancherel identity (2.9) that

$$\lambda^2 d_{Ax}(\lambda) \leq \|Ax\|^2_{\ell^2(\mathbb{Z}^d,\mu)} = \sum_{n\in\mathbb{Z}^d} |\widehat{x}(n)|^2 \overset{(2.9)}{=} \|x\|^2_{L^2(\mathbb{T}^d_\theta)}.$$

Thus, A is of weak type $(2, 2)$ with operator norm at most $c_2 = 1$. Now, let us show (4.4). Applying formulas (2.10) and (2.6), and the non-commutative Hölder inequality, we obtain

$$\frac{|\widehat{x}(n)|}{h(n)} \overset{(2.6)}{=} \frac{|\tau_\theta(x(e^\theta_n)^*)|}{h(n)} \leq \frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}\|(e^\theta_n)^*\|_{L^\infty(\mathbb{T}^d_\theta)}}{h(n)} \overset{(2.10)}{=} \frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}}{h(n)}, \quad n \in \mathbb{Z}^d.$$

Therefore, we have

$$\{n \in \mathbb{Z}^d : \frac{|\widehat{x}(n)|}{h(n)} > \lambda\} \subset \{n \in \mathbb{Z}^d : \frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}}{h(n)} > \lambda\}$$

for any $\lambda>0$. Consequently,

$$\mu\{n\in\mathbb{Z}^d:\frac{|\widehat{x}(n)|}{h(n)}>\lambda\}\leq\mu\{n\in\mathbb{Z}^d:\frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}}{h(n)}>\lambda\}$$

for any $\lambda>0$. Setting $v:=\frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}}{\lambda}$, we obtain

$$\mu\{n\in\mathbb{Z}^d:\frac{|\widehat{x}(n)|}{h(n)}>\lambda\}\leq\mu\{n\in\mathbb{Z}^d:\frac{\|x\|_{L^1(\mathbb{T}^d_\theta)}}{h(n)}>\lambda\}=\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\leq v}}h^2(n). \tag{4.5}$$

Let us estimate the right hand side. We now claim that

$$\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\leq v}}h^2(n)\leq 2v\cdot M_h. \tag{4.6}$$

Indeed, first we have

$$\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\leq v}}h^2(n)=\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\leq v}}\int\limits_0^{h^2(n)}dt.$$

By interchanging the order of integration we obtain

$$\sum_{\substack{n\in\mathbb{Z}^d\\ h(n)\leq v}}\int\limits_0^{h^2(n)}dt=\int\limits_0^{v^2}dt\sum_{\substack{n\in\mathbb{Z}^d\\ t^{\frac{1}{2}}\leq h(n)\leq v}}1.$$

Further, by a substitution $t=s^2$ we have

$$\int\limits_0^{v^2}dt\sum_{\substack{n\in\mathbb{Z}^d\\ t^{\frac{1}{2}}\leq h(n)\leq v}}1=2\int\limits_0^{v}sds\sum_{\substack{n\in\mathbb{Z}^d\\ s\leq h(n)\leq v}}1\leq 2\int\limits_0^{v}sds\sum_{\substack{n\in\mathbb{Z}^d\\ s\leq h(n)}}1.$$

Since

$$s\sum_{\substack{n\in\mathbb{Z}^d\\ s\leq h(n)}}1\leq\sup_{s>0}s\sum_{\substack{n\in\mathbb{Z}^d\\ s\leq h(n)}}1=M_h,$$

and $M_h<\infty$ by assumption, it follows that

$$2\int\limits_0^{v}sds\sum_{\substack{n\in\mathbb{Z}^d\\ s\leq h(n)}}1\leq 2v\cdot M_h.$$

This proves the claim (4.6). Combining (4.5) and (4.6) we obtain (4.4) which shows that A is indeed of weak type $(1,1)$ with operator norm at most $c_1=2M_h$. Then, by using Marcinkiewicz interpolation theorem (see, [8, Theorem 1.3.1]) with $p_1=1$, $p_2=2$ and $\frac{1}{p}=\frac{1-\eta}{1}+\frac{\eta}{2}$, we now obtain the inequality (4.2). This completes the proof. □

Remark 4.2. Let $\theta=0$ and $d=1$. If $h(n)=\frac{1}{1+|n|}, n\in\mathbb{Z}$, then $M_h<\infty$ in Theorem 4.1. Consequently, we obtain the classical Hardy-Littlewood inequality [56] (see also [2, Theorem 2.1]).

As a consequence of the Paley-type inequality in Theorem 4.1, we obtain the following Hardy-Littlewood type inequality on the non-commutative torus. We have very recently found after we have proved it that this inequality indeed, exists in [55, Remark 7.3] without proof. See also [1], where similar results are obtained for compact quantum groups. For the convenience of the reader we give a new and simple proof, as a consequence of Theorem 4.1.

Theorem 4.3. *(Hardy-Littlewood type inequality). Let $1 < p \leq 2$ and let $\varphi : \mathbb{Z}^d \to \mathbb{R}_+$ be a strictly positive function such that*

$$\sum_{n\in\mathbb{Z}^d} \frac{1}{\varphi^{\beta}(n)} < \infty \quad \text{for} \quad \text{some} \quad \beta > 0.$$

Then for any $x \in L^p(\mathbb{T}^d_\theta)$ we have

$$\left(\sum_{n\in\mathbb{Z}^d} |\widehat{x}(n)|^p \varphi^{\beta(p-2)}(n)\right)^{\frac{1}{p}} \leq c_p \|x\|_{L^p(\mathbb{T}^d_\theta)}, \tag{4.7}$$

where $c_p > 0$ is a constant independent of x.

Proof. By the assumption, we have

$$C := \sum_{n\in\mathbb{Z}^d} \frac{1}{\varphi^{\beta}(n)} < \infty \quad \text{for} \quad \text{some} \quad \beta > 0.$$

Thus,

$$C \geq \sum_{\substack{n\in\mathbb{Z}^d \\ \varphi^{\beta}(n)\leq \frac{1}{s}}} \frac{1}{\varphi^{\beta}(n)} \geq s \sum_{\substack{n\in\mathbb{Z}^d \\ \varphi^{\beta}(n)\leq \frac{1}{s}}} 1 = s \sum_{\substack{n\in\mathbb{Z}^d \\ s\leq \frac{1}{\varphi^{\beta}(n)}}} 1,$$

where $s > 0$. Hence, taking supremum with respect to $\lambda > 0$, we obtain

$$\sup_{s>0} s \sum_{\substack{n\in\mathbb{Z}^d \\ s\leq \frac{1}{\varphi^{\beta}(n)}}} 1 \leq C < \infty.$$

Then, applying Theorem 4.1 for the function $h(n) = \frac{1}{\varphi^{\beta}(n)}$, $n \in \mathbb{Z}^d$, we obtain the desired result. □

It can be seen that Theorem 4.3 shows a necessary condition for the operator x to belong to $L^p(\mathbb{T}^d_\theta)$ for $1 < p \leq 2$. However, this is not a sufficient condition. Thus, there is a natural question of finding a sufficient condition for x (in terms of its Fourier coefficients) to belong to $L^p(\mathbb{T}^d_\theta)$. The following result answers this question.

Theorem 4.4. *(Inverse Hardy-Littlewood type inequality). Let $2 \leq p < \infty$ with $\frac{1}{p} + \frac{1}{p'} = 1$ and let $\varphi : \mathbb{Z}^d \to \mathbb{R}_+$ be a strictly positive function such that*

$$\sum_{n\in\mathbb{Z}^d} \frac{1}{\varphi^{\beta}(n)} < \infty \quad \text{for} \quad \text{some} \quad \beta > 0.$$

If

$$\sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta p(2-p')}{p'}}(n) |\widehat{x}(n)|^p < \infty,$$

then $x \in L^p(\mathbb{T}^d_\theta)$ and we have

$$\|x\|_{L^p(\mathbb{T}^d_\theta)} \le C_p \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta p(2-p')}{p'}}(n) |\widehat{x}(n)|^p \right)^{\frac{1}{p}}, \tag{4.8}$$

where $C_p > 0$ is a constant independent of x.

Proof. By duality of $L^p(\mathbb{T}^d_\theta)$ we have

$$\|x\|_{L^p(\mathbb{T}^d_\theta)} = \sup\{|\tau_\theta(xy^*)| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1\}.$$

By the Plancherel (or Parseval) identity and polarization identity, we have

$$\tau_\theta(xy^*) = \sum_{n\in\mathbb{Z}^d} \widehat{x}(n)\overline{\widehat{y}(n)}, \ \ x \in L^p(\mathbb{T}^d_\theta), \ \ y \in L^{p'}(\mathbb{T}^d_\theta). \tag{4.9}$$

Since $|\widehat{x}(n)\overline{\widehat{y}(n)}| \le |\widehat{x}(n)||\overline{\widehat{y}(n)}|$ for all $n \in \mathbb{Z}^d$, applying the classical Hölder inequality for any $y \in L^{p'}(\mathbb{T}^d_\theta)$ with $\|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1$, we obtain

$$\begin{aligned}
\|x\|_{L^p(\mathbb{T}^d_\theta)} &= \sup\{|\tau_\theta(xy^*)| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1\} \\
&= \sup\left\{ \left| \sum_{n\in\mathbb{Z}^d} \widehat{x}(n)\overline{\widehat{y}(n)} \right| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1 \right\} \\
&\le \sup\left\{ \sum_{n\in\mathbb{Z}^d} |\widehat{x}(n)||\overline{\widehat{y}(n)}| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1 \right\} \\
&= \sup\left\{ \sum_{n\in\mathbb{Z}^d} |\widehat{x}(n)||\widehat{y}(n)| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1 \right\} \\
&= \sup\left\{ \sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta(2-p')}{p'}}(n)|\widehat{x}(n)| \cdot \varphi^{\frac{\beta(p'-2)}{p'}}(n)|\widehat{y}(n)| : y \in L^{p'}(\mathbb{T}^d_\theta), \quad \|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1 \right\} \\
&\le \sup_{\substack{y\in L^{p'}(\mathbb{T}^d_\theta) \\ \|y\|_{L^{p'}(\mathbb{T}^d_\theta)}=1}} \left\{ \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta p(2-p')}{p'}}(n)|\widehat{x}(n)|^p \right)^{1/p} \cdot \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\beta(p'-2)}(n)|\widehat{y}(n)|^{p'} \right)^{1/p'} \right\}.
\end{aligned}$$

Now applying Theorem 4.3 with respect to p', we get

$$\begin{aligned}
\|x\|_{L^p(\mathbb{T}^d_\theta)} &\le \sup_{\substack{y\in L^{p'}(\mathbb{T}^d_\theta) \\ \|y\|_{L^{p'}(\mathbb{T}^d_\theta)}=1}} \left\{ \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta p(2-p')}{p'}}(n)|\widehat{x}(n)|^p \right)^{1/p} \cdot \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\beta(p'-2)}(n)|\widehat{y}(n)|^{p'} \right)^{1/p'} \right\} \\
&\le c_{p'} \left(\sum_{n\in\mathbb{Z}^d} \varphi^{\frac{\beta p(2-p')}{p'}}(n)|\widehat{x}(n)|^p \right)^{1/p} \cdot \sup_{\substack{y\in L^{p'}(\mathbb{T}^d_\theta) \\ \|y\|_{L^{p'}(\mathbb{T}^d_\theta)}=1}} \|y\|_{L^{p'}(\mathbb{T}^d_\theta)}.
\end{aligned}$$

Since $\|y\|_{L^{p'}(\mathbb{T}^d_\theta)} = 1$, taking $C_p = c_{p'}$, we obtain the desired result. □

Remark 4.5. If $p=2$, then both of the statements in Theorems 4.3 and 4.4 reduce to the Plancherel identity (2.9).

The following result can be inferred from [8, Corollary 5.5.2, p. 120].

Proposition 4.6. *Let $d\mu_1(n)=\mu_1(n)d\nu(n)$, $d\mu_2(n)=\mu_2(n)d\nu(n), n\in\mathbb{Z}^d$. Suppose that $1\le p,q_0,q_1<\infty$. If a continuous linear operator A admits bounded extensions $A: L^p(\mathbb{T}^d_\theta)\to \ell^{q_0}(\mathbb{Z}^d,\mu_1)$ and $A: L^p(\mathbb{T}^d_\theta)\to \ell^{q_1}(\mathbb{Z}^d,\mu_2)$, then there exists a bounded extension $A: L^p(\mathbb{T}^d_\theta)\to \ell^{q}(\mathbb{Z}^d,\mu)$ where $0<\eta<1$, $\frac{1}{q}=\frac{1-\eta}{q_0}+\frac{\eta}{q_1}$ and $\mu=\mu_1^{\frac{q(1-\eta)}{q_0}}\cdot\mu_2^{\frac{q\eta}{q_1}}$.*

By using this interpolation result between the Paley-type inequality (4.2) in Theorem 4.1 and the Hausdorff-Young inequality (3.1) in Theorem 7.2, we obtain the following result.

Theorem 4.7. *(Hausdorff–Young–Paley inequality). Let $1<p\le r\le p'<\infty$ with $\frac{1}{p}+\frac{1}{p'}=1$. Let h be given as in Theorem 4.1. Then we have*

$$\left(\sum_{n\in\mathbb{Z}^d}\left(|\widehat{x}(n)|h(n)^{\frac{1}{r}-\frac{1}{p'}}\right)^r\right)^{\frac{1}{r}}\le c_{p,r,p'}M_h^{\frac{1}{r}-\frac{1}{p'}}\|x\|_{L^p(\mathbb{T}^d_\theta)},\tag{4.10}$$

where $c_{p,r,p'}>0$ is a constant independent of x.

Proof. As in Theorem 3.1, let us consider a linear operator $Ax:=\{\widehat{x}(n)\}_{n\in\mathbb{Z}^d}$ on $L^p(\mathbb{T}^d_\theta)$. Hence, applying the Paley-type inequality (4.2) with $1<p\le 2$, we obtain

$$\left(\sum_{n\in\mathbb{Z}^d}|\widehat{x}(n)|^p h^{2-p}(n)\right)^{\frac{1}{p}}\le c_p M_h^{\frac{2-p}{p}}\|x\|_{L^p(\mathbb{T}^d_\theta)}.\tag{4.11}$$

In other words, A is bounded from $L^p(\mathbb{T}^d_\theta)$ to the weighted space $\ell^p(\mathbb{Z}^d,\mu_1)$ with the weight $\mu_1(n):=h^{2-p}(n)>0$ for $n\in\mathbb{Z}^d$. On the other hand, by the Hausdorff-Young inequality (3.1) in Theorem 3.1, we have

$$\|\widehat{x}\|_{\ell^{p'}(\mathbb{Z}^d)}\le\|x\|_{L^p(\mathbb{T}^d_\theta)},$$

where $1\le p\le 2$ with $\frac{1}{p}+\frac{1}{p'}=1$. This shows that A is bounded from $L^p(\mathbb{T}^d_\theta)$ to $\ell^{p'}(\mathbb{Z}^d,\mu_2)$, where $\mu_2(n):=1$ for all $n\in\mathbb{Z}^d$. By Proposition 4.6 we infer that $A: L^p(\mathbb{T}^d_\theta)\to\ell^r(\mathbb{Z}^d,\mu)$ is bounded for any r such that $p\le r\le p'$, where the space $\ell^r(\mathbb{Z}^d,\mu)$ is defined as the space of sequences $a=\{a(n)\}_{n\in\mathbb{Z}^d}$ with the norm

$$\|a\|_{\ell^r(\mathbb{Z}^d,\mu)}:=\left(\sum_{n\in\mathbb{Z}^d}|a(n)|^r\mu(n)\right)^{\frac{1}{r}}$$

and μ is a positive sequence over $\mathbb{Z}^d$ to be determined. Let us compute μ in our setting. Indeed, fix $\eta\in(0,1)$ such that $\frac{1}{r}=\frac{1-\eta}{p}+\frac{\eta}{p'}$. Then we have that $\eta=\frac{p-r}{r(p-2)}$ and by Proposition 4.6 with respect to $q=r$, $q_0=p$, and $q_1=p'$, we obtain

$$\mu(n)=(\mu_1(n))^{\frac{r(1-\eta)}{p}}\cdot(\mu_2(n))^{\frac{r\eta}{p'}}=(h^{2-p}(n))^{\frac{r(1-\eta)}{p}}\cdot 1^{\frac{r\eta}{p'}}=h^{1-\frac{r}{p'}}(n)$$

for all $n\in\mathbb{Z}^d$ and $\frac{2-p}{p}\cdot(1-\eta)=\frac{1}{r}-\frac{1}{p'}$. Thus,

$$\|A(x)\|_{\ell^r(\mathbb{Z}^d,\mu)}\le c_{p,r}(M_h^{\frac{2-p}{p}})^{1-\eta}\|x\|_{L^p(\mathbb{T}^d_\theta)}=M_h^{\frac{1}{r}-\frac{1}{p'}}\|x\|_{L^p(\mathbb{T}^d_\theta)},\quad x\in L^p(\mathbb{T}^d_\theta).$$

This completes the proof. □

5. Hörmander multiplier theorem on quantum tori

In this section, we are concerned with the question of what assumptions on the symbol $g:\mathbb{Z}^d\to\mathbb{C}$ and parameters $1\leq p,q\leq\infty$ guarantee that T_g (see (2.11)) is bounded from $L^p(\mathbb{T}^d_\theta)$ to $L^q(\mathbb{T}^d_\theta)$. The following result is a non-commutative analogue of the Hörmander's result [25, Theorem 1.11] (see also [18, p. 303] for discrete version of the Hörmander's theorem) on $\mathbb{T}^d_\theta$.

Theorem 5.1. *(Hörmander multiplier theorem). Let $1<p\leq 2\leq q<\infty$ and let $g:\mathbb{Z}^d\to\mathbb{C}$ be a symbol function. Then Fourier multiplier T_g has a bounded extension from $L^p(\mathbb{T}^d_\theta)$ to $L^q(\mathbb{T}^d_\theta)$, and we have*

$$\|T_g\|_{L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)}\leq c_{p,q}\sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}, \tag{5.1}$$

where $c_{p,q}>0$ is a constant independent of x.

Proof. By duality it is sufficient to study two cases: $1<p\leq q'\leq 2$ and $1<q'\leq p\leq 2$, where $1=\frac{1}{q}+\frac{1}{q'}$.

First, we consider the case $1<p\leq q'\leq 2$, where $1=\frac{1}{q}+\frac{1}{q'}$. By (2.5) and (2.8) and (2.11) we have

$$\begin{aligned}\widehat{T_g(x)}(n)&\overset{(2.8)}{=}\tau_\theta\left(T_g(x)(e_n^\theta)^*\right)\\ &\overset{(2.11)}{=}\tau_\theta\left(\sum_{m\in\mathbb{Z}^d}g(m)\widehat{x}(m)e_m^\theta(e_n^\theta)^*\right)\\ &=\sum_{m\in\mathbb{Z}^d}g(m)\widehat{x}(m)\tau_\theta\left(e_m^\theta(e_n^\theta)^*\right)\\ &\overset{(2.5)}{=}g(n)\widehat{x}(n),\quad\forall n\in\mathbb{Z}^d.\end{aligned} \tag{5.2}$$

Then, it follows from Theorem 3.3 and (5.2) that

$$\|T_g(x)\|_{L^q\left(\mathbb{T}^d_\theta\right)}\overset{(3.2)}{\leq}\|\widehat{T_g(x)}\|_{\ell^{q'}\left(\mathbb{Z}^d\right)}\overset{(5.2)}{=}\|\{g(n)\widehat{x}(n)\}\|_{\ell^{q'}\left(\mathbb{Z}^d\right)}, \tag{5.3}$$

for all $x\in L^p(\mathbb{T}^d_\theta)$.

Therefore, if we set $q':=r$ and $\frac{1}{s}:=\frac{1}{p}-\frac{1}{q}=\frac{1}{q'}-\frac{1}{p'}$, then for $h(n):=|g(n)|^s, n\in\mathbb{Z}^d$, we are in a position to apply the Hausdorff-Young-Paley inequality in Theorem 4.7. In other words, we obtain

$$\left(\sum_{n\in\mathbb{Z}^d}\left(|\widehat{x}(n)|\cdot|g(n)|\right)^{q'}\right)^{\frac{1}{q'}}\lesssim M_{|g|^s}^{\frac{1}{s}}\|x\|_{L^p\left(\mathbb{T}^d_\theta\right)} \tag{5.4}$$

for any $x \in L^p(\mathbb{T}^d_\theta)$. Let us study $M^{\frac{1}{s}}_{|g|^s}$ separately. Indeed, by definition

$$M^{\frac{1}{s}}_{|g|^s} := \left(\sup_{\lambda>0} \lambda \sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)|^s \geq \lambda}} 1 \right)^{\frac{1}{s}} = \left(\sup_{\lambda>0} \lambda \sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda^{\frac{1}{s}}}} 1 \right)^{\frac{1}{s}} = \left(\sup_{\lambda>0} \lambda^s \sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda}} 1 \right)^{\frac{1}{s}}.$$

Since $\frac{1}{s} := \frac{1}{p} - \frac{1}{q}$, it follows that

$$\begin{aligned} M^{\frac{1}{s}}_{|g|^s} &= \left(\sup_{\lambda>0} \lambda^s \sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda}} 1 \right)^{\frac{1}{p}-\frac{1}{q}} \\ &= \sup_{\lambda>0} \lambda^{s\left(\frac{1}{p}-\frac{1}{q}\right)} \left(\sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda}} 1 \right)^{\frac{1}{p}-\frac{1}{q}} \\ &= \sup_{\lambda>0} \lambda \left(\sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda}} 1 \right)^{\frac{1}{p}-\frac{1}{q}}. \end{aligned} \tag{5.5}$$

Hence, combining (5.3), (5.4), and (5.5) we obtain

$$\begin{aligned} \|T_g(x)\|_{L^q(\mathbb{T}^d_\theta)} &\overset{(5.3)}{\lesssim} \left(\sum_{n\in\mathbb{Z}^d} \left(|\widehat{x}(n)| \cdot |g(n)| \right)^{q'} \right)^{\frac{1}{q'}} \\ &\overset{(5.4)}{\lesssim} M^{\frac{1}{s}}_{|g|^s} \|x\|_{L^p\left(\mathbb{T}^d_\theta\right)} \\ &\overset{(5.5)}{=} \sup_{\lambda>0} \lambda \left(\sum_{\substack{n\in\mathbb{Z}^d \\ |g(n)| \geq \lambda}} 1 \right)^{\frac{1}{p}-\frac{1}{q}} \|x\|_{L^p(\mathbb{T}^d_\theta)}, \end{aligned}$$

for $1 < p \leq q' \leq 2$ and $x \in L^p(\mathbb{T}^d_\theta)$.

Next, we consider the case $q' \leq p \leq 2$ so that $p' \leq (q')' = q$, where $1 = \frac{1}{q} + \frac{1}{q'}$ and $1 = \frac{1}{p} + \frac{1}{p'}$. Thus, the L^p-duality (see Lemma 2.3) yields that $T^*_g = T_{\overline{g}}$ and

$$\|T_g\|_{L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)} = \|T_{\overline{g}}\|_{L^{q'}(\mathbb{T}^d_\theta)\to L^{p'}(\mathbb{T}^d_\theta)}.$$

The symbol of the adjoint operator T^*_g equals to $\overline{g}$ and obviously we have $|g(n)| = |\overline{g(n)}|$ for all $n \in \mathbb{Z}^d$. Set $\frac{1}{p} - \frac{1}{q} = \frac{1}{s} = \frac{1}{q'} - \frac{1}{p'}$. Hence, by repeating the argument in the previous case

we have

$$\begin{aligned}\|T_{\overline{g}}(x)\|_{L^{p'}(\mathbb{T}^d_\theta)} &\lesssim \sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |\overline{g(n)}|\geq\lambda}}1\right)^{\frac{1}{q'}-\frac{1}{p'}}\|x\|_{L^{q'}(\mathbb{T}^d_\theta)}\\ &= \sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}\|x\|_{L^{q'}(\mathbb{T}^d_\theta)}.\end{aligned}$$

In other words, we have

$$\|T_{\overline{g}}\|_{L^{q'}(\mathbb{T}^d_\theta)\to L^{p'}(\mathbb{T}^d_\theta)}\lesssim \sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}.$$

Combining both cases, we obtain

$$\|T_g\|_{L^{p}(\mathbb{T}^d_\theta)\to L^{q}(\mathbb{T}^d_\theta)}\lesssim \sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}$$

for all $1<p\leq 2\leq q<\infty$. This concludes the result. □

As an application of Theorem 5.1 we will show the L^p-L^q estimate for the heat semigroup on quantum tori. Define the partial differentiation operators ∂_j^θ, $j=1,...,d$, by the formula

$$\partial_j^\theta(e_n^\theta)=n_je_n^\theta,\quad n=(n_1,...,n_d)\in\mathbb{Z}^d. \tag{5.6}$$

Every partial derivation ∂_j^θ can be viewed as a densely defined closed (unbounded) operator on $L^2(\mathbb{T}^d_\theta)$, whose adjoint is equal to $-\partial_j^\theta$. Let $\Delta_\theta=(\partial_1^\theta)^2+\cdots+(\partial_d^\theta)^2$ be the Laplacian. Then $\Delta_\theta=-((\partial_1^\theta)^*\partial_1^\theta+\cdots+(\partial_d^\theta)^*\partial_d^\theta)$, so $-\Delta_\theta$ is a positive operator on $L^2(\mathbb{T}^d_\theta)$ with spectrum equal to $\{|n|^2:n\in\mathbb{Z}^d\}$ (see [34],[54]). Let us define the heat semigroups of operators on $L^2(\mathbb{T}^d_\theta)$ by $t\mapsto e^{t\Delta_\theta}$. This operator can be defined as the Fourier multiplier. Observe that by definition (see (2.11)), it is an operator of the form T_g with the symbol $g(n)=e^{-t|n|^2}$, $n\in\mathbb{Z}^d$, $t>0$. Then a simple application of Theorem 5.1 gives us the L^p-L^q properties of the propagator and the time asymptotics of its non-commutative Lorentz space norm.

Corollary 5.2. *Let $1<p\leq 2\leq q<\infty$. We have*

$$\|e^{t\Delta_\theta}\|_{L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)}\leq c_{p,q}t^{-\frac{d}{2}\left(\frac{1}{p}-\frac{1}{q}\right)},\quad t>0.$$

Proof. By Theorem 5.1 we have

$$\|e^{t\Delta_\theta}\|_{L^p(\mathbb{T}^d_\theta)\to L^q(\mathbb{T}^d_\theta)}\leq c_{p,q}\cdot\sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ e^{-t|n|^2}\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}.$$

Then a simple calculation shows that the right hand side is less or equal than the function $t^{-\frac{d}{2}\left(\frac{1}{p}-\frac{1}{q}\right)}$, $t>0$. This completes the proof. □

6. Sobolev and logarithmic Sobolev inequalities on quantum tori

In this section, we obtain Sobolev and logarithmic Sobolev type inequalities on quantum tori. Let $\partial_j^\theta, j=1,\cdots,d$, be differentiation operators defined by (5.6) in the previous section. Then, every partial derivation ∂_j^θ can be viewed as a densely defined closed (unbounded) operator on $L^2(\mathbb{T}_\theta^d)$, whose adjoint is equal to $-\partial_j^\theta$. Similarly, we define the gradient $\nabla_{\mathbb{T}_\theta^d}$ by the formula

$$\nabla_\theta = (\partial_1^\theta, \cdots, \partial_d^\theta),$$

which may be considered as a linear operator on $L^2(\mathbb{T}_\theta^d)$. For a multi-index $\alpha=(\alpha_1,...,\alpha_d)$, define

$$\nabla_\theta^\alpha = (\partial_1^\theta)^{\alpha_1}\cdots(\partial_d^\theta)^{\alpha_d},$$

which is also considered as a linear operator on $L^2(\mathbb{T}_\theta^d)$.

Definition 6.1. (see, [54, Definition 2.6. (ii)]) Let $s\in\mathbb{R}$ and $1\le p<\infty$. Then the potential (or fractional) Sobolev space of order s is defined to be

$$L_s^p(\mathbb{T}_\theta^d) = \{x\in\mathcal{D}'(\mathbb{T}_\theta^d) : (1-\Delta_\theta)^{\frac{s}{2}}x\in L^p(\mathbb{T}_\theta^d)\},$$

equipped with the norm

$$\|x\|_{L_s^p(\mathbb{T}_\theta^d)} = \|(1-\Delta_\theta)^{\frac{s}{2}}x\|_{L^p(\mathbb{T}_\theta^d)},$$

where

$$\Delta_\theta = (\partial_1^\theta)^2+\cdots+(\partial_d^\theta)^2 = -\nabla_\theta^*\nabla_\theta \tag{6.1}$$

is the Laplacian. If $p=2$, then as usual, we denote $L_s^2(\mathbb{T}_\theta^d) =: H^s(\mathbb{T}_\theta^d)$. If $s:=k\in\mathbb{N}$, then the corresponding Sobolev space $W^{k,p}(\mathbb{T}_\theta^d)$ is defined to be the set of $x\in L^p(\mathbb{T}_\theta^d)$ with $\nabla_\theta^\alpha x\in L^p(\mathbb{T}_\theta^d)$ for all $|\alpha|_1\le k$, where $|\alpha|_1=\alpha_1+...+\alpha_d$. The $W^{k,p}$ norm is the sum of the L^p norms of $\nabla_\theta^\alpha x$ for all $0\le|\alpha|_1\le k$, that is

$$\|x\|_{W^{k,p}(\mathbb{T}_\theta^d)} = \left(\sum_{0\le|\alpha|_1\le k}\|\nabla_\theta^\alpha x\|_{L^p(\mathbb{T}_\theta^d)}^p\right)^{1/p}.$$

The above definitions of Sobolev spaces on quantum tori and their main properties including embeddings were studied in [54] (see also [52] for the isomorphism property between L^p and Sobolev spaces when $s\in\mathbb{N}$).

As another consequence of Theorem 5.1 we obtain the following Sobolev type embedding.

Theorem 6.2. *Let $1<p\le 2\le q<\infty$. Then the inequality*

$$\|x\|_{L^q(\mathbb{T}_\theta^d)} \le c_{p,q,s}\|x\|_{L_s^p(\mathbb{T}_\theta^d)},$$

where $c_{p,q,s}>0$ is a constant independent of x, holds true provided that $s\ge d\left(\frac{1}{p}-\frac{1}{q}\right)$.

Proof. Let us consider $g(n)=(1+|n|^2)^{-s/2}, \quad n\in\mathbb{Z}^d$, and $s>0$. Take $T_g=(1-\Delta_\theta)^{-s/2}$. In order to use Theorem 5.1, we need first to calculate the right hand side of (5.1). For this, we have

$$\sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}} = \sup_{\lambda>0}\lambda\left(\sharp\{n\in\mathbb{Z}^d:\frac{1}{(1+|n|^2)^{-s/2}}\geq\lambda\}\right)^{\frac{1}{p}-\frac{1}{q}}.$$

Since

$$\sharp\{n\in\mathbb{Z}^d:\frac{1}{(1+|n|^2)^{-s/2}}\geq\lambda\}=\begin{cases}0, & \text{if } 1<\lambda<\infty,\\ \sharp\{n\in\mathbb{Z}^d:|n|\leq\sqrt{\lambda^{-2/s}-1}\}, & \text{if } 0<\lambda\leq 1,\end{cases}$$

it follows that

$$\begin{aligned}\sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}} &= \sup_{0<\lambda\leq 1}\lambda\left(\sharp\{n\in\mathbb{Z}^d:\frac{1}{(1+|n|^2)^{-s/2}}\geq\lambda\}\right)^{\frac{1}{p}-\frac{1}{q}}\\ &= \sup_{0<\lambda\leq 1}\lambda\left(\sharp\{n\in\mathbb{Z}^d:|n|\leq\sqrt{\lambda^{-2/s}-1}\}\right)^{\frac{1}{p}-\frac{1}{q}}\\ &\leq \sup_{0<\lambda\leq 1}\lambda\left(\sharp\{n\in\mathbb{Z}^d:|n|\leq\lambda^{-1/s}\}\right)^{\frac{1}{p}-\frac{1}{q}}\\ &\lesssim \sup_{0<\lambda\leq 1}\lambda^{1-\frac{d}{s}(\frac{1}{p}-\frac{1}{q})}<\infty,\end{aligned}$$

whenever $s\geq d\left(\frac{1}{p}-\frac{1}{q}\right)$, where $\sharp$ is a counting measure on $\mathbb{Z}^d$. Thus, the function g satisfies the assumptions of Theorem 5.1, and consequently, we have

$$\begin{aligned}\|x\|_{L^q(\mathbb{T}^d_\theta)} &= \|(1-\Delta_\theta)^{-s/2}(1-\Delta_\theta)^{s/2}x\|_{L^p(\mathbb{T}^d_\theta)}\\ &\overset{(5.1)}{\leq} c_{p,q,s}\sup_{\lambda>0}\lambda\left(\sum_{\substack{n\in\mathbb{Z}^d\\ |g(n)|\geq\lambda}}1\right)^{\frac{1}{p}-\frac{1}{q}}\|(1-\Delta_\theta)^{s/2}x\|_{L^p(\mathbb{T}^d_\theta)}\\ &\leq c_{p,q,s}\|x\|_{L^p_s(\mathbb{T}^d_\theta)}\end{aligned}$$

for $s\geq d\left(\frac{1}{p}-\frac{1}{q}\right)$ and $x\in L^p_s(\mathbb{T}^d_\theta)$, thereby completing the proof. □

We also obtain the following a simple consequence of the non-commutative Hölder inequality.

Lemma 6.3. *Let $1\leq p\leq r\leq q\leq\infty$ be such that $\frac{1}{r}=\frac{\eta}{p}+\frac{1-\eta}{q}$ for some $\eta\in(0,1)$. Then for any $x\in L^q(\mathbb{T}^d_\theta)$, we have*

$$\|x\|_{L^r(\mathbb{T}^d_\theta)}\leq\|x\|^{\eta}_{L^p(\mathbb{T}^d_\theta)}\|x\|^{1-\eta}_{L^q(\mathbb{T}^d_\theta)}. \tag{6.2}$$

Proof. The proof is a simple consequence of the non-commutative Hölder inequality [53, Theorem 5.2.9], therefore, is omitted. □

Lemma 6.4. *(Logarithmic Hölder inequality) Let $1 \leq p < q < \infty$. Then for any $0 \neq x \in L^p(\mathbb{T}^d_\theta)$ we have*

$$\tau_\theta\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\log\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\right)\right) \leq \frac{q}{q-p}\log\left(\frac{\|x\|^p_{L^q(\mathbb{T}^d_\theta)}}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\right), \tag{6.3}$$

where $Spec(x)$ is the spectrum of x.

Proof. The proof is a modification of the proof in [14]. Let us consider the function

$$f(\frac{1}{r}) = \log(\|x\|_{L^r(\mathbb{T}^d_\theta)}), \quad r > 0.$$

By Lemma 6.3 we have

$$\begin{aligned} f(\frac{1}{r}) &= \log(\|x\|_{L^r(\mathbb{T}^d_\theta)}) \leq \log(\|x\|^\eta_{L^p(\mathbb{T}^d_\theta)}\|x\|^{1-\eta}_{L^q(\mathbb{T}^d_\theta)}) \\ &= \log(\|x\|^\eta_{L^p(\mathbb{T}^d_\theta)}) + \log(\|x\|^{1-\eta}_{L^q(\mathbb{T}^d_\theta)}) \\ &= \eta f(\frac{1}{p}) + (1-\eta)f(\frac{1}{q}), \end{aligned}$$

with $r > 1$ and $\eta \in (0,1)$ such that $\frac{1}{r} = \frac{\eta}{p} + \frac{1-\eta}{q}$. This shows that the function f is convex. Since

$$f(s) = s\log(\tau_\theta(|x|^{1/s})),$$

taking derivative with respect to s, we obtain

$$\begin{aligned} f'(s) &= \log(\tau_\theta(|x|^{1/s})) + s\Big(\log(\tau_\theta(|x|^{1/s}))\Big)'_s \\ &= \log(\tau_\theta(|x|^{1/s})) + s\frac{\Big(\tau_\theta(|x|^{1/s})\Big)'_s}{\tau_\theta(|x|^{1/s})} \\ &= \log(\tau_\theta(|x|^{1/s})) - \frac{1}{s}\frac{\tau_\theta(|x|^{1/s}\log(|x|))}{\tau_\theta(|x|^{1/s})}. \end{aligned}$$

On the other hand, since f is a differentiable convex function of s, it follows that

$$f'(s) \geq \frac{f(s_0)-f(s)}{s_0-s}, \quad s_0 > s > 0.$$

Taking $s = \frac{1}{p}$ and $s_0 = \frac{1}{q}$, we have

$$p\frac{\tau_\theta(|x|^p\log(|x|))}{\tau_\theta(|x|^p)} - \log(\tau_\theta(|x|^p)) \leq \frac{pq}{q-p}\log\left(\frac{\|x\|_{L^q(\mathbb{T}^d_\theta)}}{\|x\|_{L^p(\mathbb{T}^d_\theta)}}\right). \tag{6.4}$$

But, by linearity of τ_θ the left hand side of the previous inequality is represented as follows

$$\begin{aligned} p & \frac{\tau_\theta(|x|^p\log(|x|))}{\tau_\theta(|x|^p)} - \log(\tau_\theta(|x|^p)) = p\frac{\tau_\theta(|x|^p\log(|x|))}{\tau_\theta(|x|^p)} - \frac{\tau_\theta(|x|^p)\cdot\log(\tau_\theta(|x|^p))}{\tau_\theta(|x|^p)} \\ &= \frac{\tau_\theta(|x|^p\log(|x|^p))}{\tau_\theta(|x|^p)} - \frac{\tau_\theta\Big(|x|^p\log(\|x\|^p_{L^p(\mathbb{T}^d_\theta)})\Big)}{\tau_\theta(|x|^p)} \\ &= \frac{\tau_\theta(|x|^p\log(|x|^p)) - \tau_\theta\Big(|x|^p\log(\|x\|^p_{L^p(\mathbb{T}^d_\theta)})\Big)}{\tau_\theta(|x|^p)} \\ &= \frac{\tau_\theta\Big(|x|^p\Big(\log(|x|^p) - \log(\|x\|^p_{L^p(\mathbb{T}^d_\theta)})\Big)\Big)}{\tau_\theta(|x|^p)} = \tau_\theta\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\log\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\right)\right). \end{aligned}$$

Hence, the assertion follows from (6.4). □

As an application of Theorem 6.2 for the case $1 < p \le 2 \le p^* < \infty$ we obtain the following logarithmic type Sobolev inequality. Another type of logarithmic Sobolev inequality was obtained recently by [28] on quantum tori when $p = 2$.

Theorem 6.5. *(Logarithmic Sobolev inequality) Let $s > 0$ and $1 < p < \infty$ be such that $d > sp$. Then for any $0 \neq x \in L^p_s(\mathbb{T}^d_\theta)$ we have the fractional logarithmic Sobolev inequality*

$$\tau_\theta\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\log\left(\frac{|x|^p}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\right)\right) \le \frac{d}{sp}\log\left(C_{p,d}\frac{\|x\|^p_{L^p_s(\mathbb{T}^d_\theta)}}{\|x\|^p_{L^p(\mathbb{T}^d_\theta)}}\right), \tag{6.5}$$

where $C_{p,d} > 0$ is a constant independent of x.

Proof. By Theorem 6.2 for $1 < p \le 2 \le p^* < \infty$ and by [54, Theorem 6.6. (i), p. 70] for other cases, we have $L^p_s(\mathbb{T}^d_\theta) \subset L^{p^*}(\mathbb{T}^d_\theta)$ with $1 < p < p^* < \infty$ and $s > 0$ such that $s = \frac{d}{p} - \frac{d}{p^*}$. This means that there exists a constant $C_{p,p^*,d} > 0$ such that

$$\|x\|_{L^{p^*}(\mathbb{T}^d_\theta)} \le C_{p,p^*,d}\|x\|_{L^p_s(\mathbb{T}^d_\theta)}, \quad x \in L^p_s(\mathbb{T}^d_\theta).$$

Taking $q := p^* = \frac{dp}{d-sp}$ in Lemma 6.4 and since $\frac{p^*}{p^*-q} = \frac{d}{sp}$, the assertion follows from (6.3). □

6.1. **Nash type inequality on quantum tori.** In this subsection, we prove the Nash inequality on quantum tori. In the classical case, this inequality is applied as a main tool in computing the decay rate for the heat equation for the sub-Laplacian.

Theorem 6.6. *(Nash type inequality). Let $x \in W^{1,2}(\mathbb{T}^d_\theta)$ and $d > 2$. Then we have*

$$\|x\|^{1+\frac{2}{d}}_{L^2(\mathbb{T}^d_\theta)} \le C_{1,d}\|x\|_{W^{1,2}(\mathbb{T}^d_\theta)}\|x\|^{\frac{2}{d}}_{L^1(\mathbb{T}^d_\theta)}, \tag{6.6}$$

where $C_{1,d} > 0$ is a constant independent of x.

Proof. By the Sobolev inequality in Theorem 6.2, we get

$$\|x\|_{L^{2^*}(\mathbb{T}^d_\theta)} \le C_{1,d}\|x\|_{W^{1,2}(\mathbb{T}^d_\theta)}, \tag{6.7}$$

where $C_{1,d} > 0$ is a constant independent of x.

By Lemma 6.3 with $r=2$, $q=2^*=\frac{2d}{d-2}$, and $p=1$, we obtain

$$\|x\|_{L^2(\mathbb{T}^d_\theta)} \leq \|x\|^{\eta}_{L^1(\mathbb{T}^d_\theta)} \|x\|^{1-\eta}_{L^{2^*}(\mathbb{T}^d_\theta)}, \tag{6.8}$$

where

$$\eta := \frac{\frac{1}{r}-\frac{1}{q}}{\frac{1}{p}-\frac{1}{q}} = \frac{\frac{1}{2}-\frac{d-2}{2d}}{\frac{d+2}{2d}} = \frac{2}{d+2}. \tag{6.9}$$

Hence, combining (6.7), (6.8), and (6.9), we get

$$\|x\|_{L^2(\mathbb{T}^d_\theta)} \overset{(6.7)\,(6.8)}{\leq} (C_{1,d}\|x\|_{W^{1,2}(\mathbb{T}^d_\theta)})^{1-\eta}\|x\|^{\eta}_{L^1(\mathbb{T}^d_\theta)} \overset{(6.9)}{=} C_{1,d}^{\frac{d}{d+2}}\|x\|^{\frac{d}{d+2}}_{W^{1,2}(\mathbb{T}^d_\theta)}\|x\|^{\frac{2}{d+2}}_{L^1(\mathbb{T}^d_\theta)}.$$

Rising both sides of the last inequality to the power $\frac{d+2}{d}$, one gets inequality (6.6). □

Remark 6.7. In fact, with our approach we can prove the extended version of the Nash inequality in Theorem 6.6. Note that it is well-known in Euclidean spaces that the logarithmic Sobolev inequality is equivalent to the Nash inequality when $p=2$. Moreover, as an application of the Nash inequality, one can compute the decay rate for the heat equation for the sub-Laplacian. For more details about these inequalities and their applications in PDEs for Lie groups we refer the reader to [14].

The following result shows an application of the Nash inequality to compute the decay rate for the heat equation with $\Delta_\theta = -\nabla^*_\theta\nabla_\theta$ as in (6.1).

Corollary 6.8. *Let $u_0 \in L^2(\mathbb{T}^d_\theta)$ be a self-adjoint operator such that*

$$\partial_t u(t) = \Delta_\theta u(t) - u(t), \quad u(0)=u_0. \tag{6.10}$$

Then the solution of equation (6.10) has the following estimate

$$\|u(t)\|_{L^2(\mathbb{T}^d_\theta)} \leq \left(1+\frac{4}{dC^2_{1,d}}t\right)^{-\frac{d}{4}}\|u_0\|_{L^2(\mathbb{T}^d_\theta)},$$

for all $t\geq 0$, where $C_{1,d}>0$ is the constant in (6.6).

Proof. If $u_0 = \sum\limits_{n\in\mathbb{Z}^d}\widehat{u_0}(n)e^\theta_n$ and $u(t)=\sum\limits_{n\in\mathbb{Z}^d}\widehat{u}(n)e^\theta_n$, then the solution of equation (6.10) is written as

$$u(t) = \sum_{n\in\mathbb{Z}^d} e^{-t(|n|^2+1)}\widehat{u_0}(n)e^\theta_n, \tag{6.11}$$

where $\widehat{u_0}(n)=\tau_\theta(u_0(e^\theta_n)^*)$, $n\in\mathbb{Z}^d$. Now, multiplying by u^* the heat equation (6.10) and taking the trace τ_θ we obtain

$$\tau_\theta((\partial_t u)u^*) = -\tau_\theta((-\Delta_\theta u)u^*) - \tau_\theta(uu^*). \tag{6.12}$$

By a direct computation, the right hand side is

$$\begin{aligned}
-\tau_\theta((-\Delta_\theta u)u^*)-\tau_\theta(uu^*) &= -\sum_{n\in\mathbb{Z}^d}\widehat{-\Delta_\theta u}(n)\cdot\overline{\widehat{u}(n)}-\tau_\theta(|u^*|^2)\\
&= -\sum_{n\in\mathbb{Z}^d}|n|^2\widehat{u}(n)\cdot\overline{\widehat{u}(n)}-\|u^*\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&= -\sum_{n\in\mathbb{Z}^d}|n|\widehat{u}(n)\cdot\overline{|n|\widehat{u}(n)}-\|u\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&= -\sum_{n\in\mathbb{Z}^d}\widehat{\nabla_\theta u}(n)\cdot\overline{\widehat{\nabla_\theta u}(n)}-\|u\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&\overset{(4.9)}{=} -\tau_\theta(\nabla_\theta(u)(\nabla_\theta u)^*)-\|u\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&= -\|(\nabla_\theta u)^*\|^2_{L^2(\mathbb{T}^d_\theta)}-\|u\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&= -\|\nabla_\theta u\|^2_{L^2(\mathbb{T}^d_\theta)}-\|u\|^2_{L^2(\mathbb{T}^d_\theta)}. \qquad (6.13)
\end{aligned}$$

Since u_0 is a self-adjoint operator, we have

$$\begin{aligned}
\overline{\widehat{u_0}(n)}(e^\theta_n)^* &= \overline{\tau_\theta(u_0(e^\theta_n)^*)}(e^\theta_n)^* \overset{(2.2)}{=} \tau_\theta(e^\theta_n u_0)e^{-i\sum\limits_{j<k}\theta_{jk}n_jn_k}e^\theta_{-n}\\
&= \tau_\theta(u_0e^\theta_n)e^{-i\sum\limits_{j<k}\theta_{jk}n_jn_k}e^\theta_{-n}\\
&= \tau_\theta(u_0e^{-i\sum\limits_{j<k}\theta_{jk}(-n_j)(-n_k)}e^\theta_n)e^\theta_{-n}\\
&\overset{(2.2)}{=} \tau_\theta(u_0(e^\theta_{-n})^*)e^\theta_{-n}=\widehat{u_0}(-n)e^\theta_{-n},\quad n\in\mathbb{Z}^d.
\end{aligned}$$

It follows that the solution (6.11) is self-adjoint. Indeed, we have

$$\begin{aligned}
u^*(t) &\overset{(6.11)}{=} \sum_{n\in\mathbb{Z}^d}e^{-t(|n|^2+1)}\overline{\widehat{u_0}(n)}(e^\theta_n)^*=\sum_{n\in\mathbb{Z}^d}e^{-t(|n|^2+1)}\widehat{u_0}(-n)e^\theta_{-n}\\
&\overset{n=-n}{=} \sum_{n\in\mathbb{Z}^d}e^{-t(|n|^2+1)}\widehat{u_0}(n)e^\theta_n=u(t).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{d}{dt}\|u(t)\|^2_{L^2(\mathbb{T}^d_\theta)} &= \tau_\theta((\partial_t u)u)+\tau_\theta(u(\partial_t u))\\
&= 2\tau_\theta((\partial_t u)u)\\
&= -2\tau_\theta((-\Delta_\theta u)u)-2\tau_\theta(|u|^2)\\
&= -2\|\nabla_\theta u\|^2_{L^2(\mathbb{T}^d_\theta)}-2\|u\|^2_{L^2(\mathbb{T}^d_\theta)}\\
&\asymp -2\|u\|^2_{W^{1,2}(\mathbb{T}^d_\theta)}.
\end{aligned}$$

Since $\widehat{u}(n)=e^{-t(1+|n|^2)}\widehat{u_0}(n), n\in\mathbb{Z}^d$, by the Plancherel identity (2.9), we have

$$\|u(t)\|^2_{L^2(\mathbb{T}^d_\theta)}\overset{(2.9)}{=}\sum_{n\in\mathbb{Z}^d}e^{-2t(1+|n|^2)}|\widehat{u_0}(n)|^2\le\sum_{n\in\mathbb{Z}^d}|\widehat{u_0}(n)|^2\overset{(2.9)}{=}\|u_0\|^2_{L^2(\mathbb{T}^d_\theta)},$$

which implies that

$$\|u(t)\|_{L^2(\mathbb{T}^d_\theta)}\le\|u_0\|_{L^2(\mathbb{T}^d_\theta)}. \qquad (6.14)$$

If we denote by $y(t) := \|u(t)\|^2_{L^2(\mathbb{T}^d_\theta)}$, then applying the Nash inequality (6.6), and by (6.14) we obtain

$$y' \overset{(6.6)}{\leq} -2C_{1,d}^{-2}\|u\|^{-\frac{4}{d}}_{L^1(\mathbb{T}^d_\theta)}y^{1+\frac{2}{d}} \leq -2C_{1,d}^{-2}\|u\|^{-\frac{4}{d}}_{L^2(\mathbb{T}^d_\theta)}y^{1+\frac{2}{d}} \overset{(6.14)}{\leq} -2C_{1,d}^{-2}\|u_0\|^{-\frac{4}{d}}_{L^2(\mathbb{T}^d_\theta)}y^{1+\frac{2}{d}}.$$

Hence, integrating with respect to $t \geq 0$, we obtain the estimate for the solution of (6.10)

$$\|u(t)\|_{L^2(\mathbb{T}^d_\theta)} \leq \left(1 + \frac{4}{dC_{1,d}^2}t\right)^{-\frac{d}{4}} \|u_0\|_{L^2(\mathbb{T}^d_\theta)},$$

which completes the proof. □

7. Nikolskii inequality

In this section, we prove Nikolskii inequality for trigonometric polynomials on the noncommutative torus. Further, we apply this inequality to study embedding properties between Besov, Wiener, and Beurling spaces.

Let Λ be a finite subset of $\mathbb{Z}^d$. Define a trigonometric polynomial x_Λ on $L^\infty(\mathbb{T}^d_\theta)$ by

$$x_\Lambda := \sum_{m\in\Lambda} \widehat{x}_\Lambda(m)e^\theta_m, \tag{7.1}$$

where $\widehat{x}_\Lambda(m) = \tau_\theta(x_\Lambda(e^\theta_m)^*)$, $m \in \Lambda$. The number of elements of $\Lambda \subset \mathbb{Z}^d$ is denoted by $N(\Lambda)$. We also define

$$\operatorname{supp}(\widehat{x}_\Lambda) := \{m \in \mathbb{Z}^d : \widehat{x}_\Lambda(m) \neq 0\}.$$

We denote by T^Λ the set of trigonometric polynomials as in (7.1). Then it is not difficult to see that $T^\Lambda \subset L^p(\mathbb{T}^d_\theta)$ for all $0 < p \leq \infty$.

Lemma 7.1. *Let x_Λ be a trigonometric polynomial on $L^\infty(\mathbb{T}^d_\theta)$ as in (7.1). Then, $x^*_\Lambda x_\Lambda$ and x^k_Λ are also trigonometric polynomials for any power $k \in \mathbb{Z}_+$, respectively.*

Proof. Using multi-index notation and (2.2) and (7.1), we can write polynomials x^*_Λ and x_Λ in the form

$$\begin{aligned}
x^*_\Lambda x_\Lambda &\overset{(7.1)}{=} \Big(\sum_{m\in\Lambda} \widehat{x}_\Lambda(m)e^\theta_m\Big)^* \sum_{n\in\Lambda} \widehat{x}_\Lambda(n)e^\theta_n \\
&= \sum_{m,n\in\Lambda} \overline{\widehat{x}_\Lambda(m)}\widehat{x}_\Lambda(n)(e^\theta_m)^*e^\theta_n \\
&\overset{(2.2)}{=} \sum_{m\in\Lambda}\Big(\sum_{n\in\Lambda} \overline{\widehat{x}_\Lambda(m)}\widehat{x}_\Lambda(n)e^{\,i\sum\limits_{j<k}\theta_{jk}m_j(n_k-m_k)}\Big)e^\theta_{n-m} \\
&= \sum_{l\in\Lambda_1}\Big(\sum_{n-m=l} \overline{\widehat{x}_\Lambda(m)}\widehat{x}_\Lambda(n)e^{\,i\sum\limits_{j<k}\theta_{jk}m_j(n_k-m_k)}\Big)e^\theta_l \\
&= \sum_{l\in\Lambda_1} \widehat{x}_{\Lambda_1}(l)e^\theta_l,
\end{aligned}$$

where $\widehat{x}_{\Lambda_1}(l) := \sum\limits_{n-m=l} \overline{\widehat{x}_\Lambda(m)}\widehat{x}_\Lambda(n)e^{\,i\sum\limits_{j<k}\theta_{jk}m_j(n_k-m_k)}$ and $\Lambda_1 := \{l = n - m : n, m \in \Lambda\}$.

We shall prove the next statement by induction. Clearly, the statement holds when $k = 2$:

$$\begin{aligned} x_\Lambda^2 &\overset{(7.1)}{=} \sum_{m,n\in\Lambda} \widehat{x}_\Lambda(m)\widehat{x}_\Lambda(n)e_m^\theta e_n^\theta \\ &\overset{(2.2)}{=} \sum_{m,n\in\Lambda} \widehat{x}_\Lambda(m)\widehat{x}_\Lambda(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)}e_{m+n}^\theta \\ &= \sum_{l\in\Lambda_2}(\sum_{n+m=l} \widehat{x}_\Lambda(m)\widehat{x}_\Lambda(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)})e_l^\theta \\ &= \sum_{l\in\Lambda_2} \widehat{x}_{\Lambda_2}(l)e_l^\theta, \end{aligned}$$

where $\widehat{x}_{\Lambda_2}(l) := \sum\limits_{n+m=l} \widehat{x}_\Lambda(m)\widehat{x}_\Lambda(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)}$ and $\Lambda_2 := \{l = n+m : n,m \in \Lambda\}$.

Suppose the statement $x_{\Lambda_\kappa}^\kappa = \sum\limits_{n\in\Lambda_\kappa} \widehat{x}_{\Lambda_\kappa}(n)e_n^\theta$ holds for any power $\kappa \in \mathbb{Z}_+$ (Λ_κ - a finite subset of $\mathbb{Z}^d$). Now, consider the case $\kappa+1$. By the above assumption, we have

$$\begin{aligned} x_\Lambda^{\kappa+1} &= x_\Lambda x_{\Lambda_\kappa}^\kappa \\ &\overset{(7.1)}{=} \sum_{m\in\Lambda} \widehat{x}_\Lambda(m)e_m^\theta \sum_{n\in\Lambda_\kappa} \widehat{x}_{\Lambda_\kappa}(n)e_n^\theta \\ &\overset{(2.2)}{=} \sum_{m\in\Lambda}\sum_{n\in\Lambda_\kappa} \widehat{x}_\Lambda(m)\widehat{x}_{\Lambda_\kappa}(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)}e_{m+n}^\theta \\ &= \sum_{s\in\Lambda_{\kappa+1}} \Big(\sum_{n+m=s} \widehat{x}_\Lambda(m)\widehat{x}_{\Lambda_\kappa}(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)}\Big)e_s^\theta \\ &= \sum_{l\in\Lambda_2} \widehat{x}_{\Lambda_{\kappa+1}}(l)e_l^\theta, \end{aligned}$$

where $\widehat{x}_{\Lambda_{\kappa+1}} := \sum\limits_{n+m=s} \widehat{x}_\Lambda(m)\widehat{x}_{\Lambda_\kappa}(n)e^{-i(\sum\limits_{j<k}\theta_{jk}m_jn_k)}$ and

$$\Lambda_{\kappa+1} := \{s = m+n : m \in \Lambda, \quad n \in \Lambda_\kappa\}.$$

This completes the proof. □

The following is an analogue of the classical Nikolskii inequality (see [49, Proposition, p.147]) on the non-commutative torus. The proof is a modification of the proof in [39, Theorem 3.1, p. 989].

Theorem 7.2. *(Nikolskii inequality). Let $0 < p \le q \le \infty$ and let γ be the smallest even number larger than or equal to $\frac{p}{2}$ for $2 < p < \infty$ and $\gamma = 1$ for $0 < p \le 2$. Let x_Λ be a trigonometric polynomial as in (7.1). Then we have*

$$\|x_\Lambda\|_{L^q(\mathbb{T}_\theta^d)} \le C_{\gamma,x_\Lambda}^{\frac{1}{p}-\frac{1}{q}} \|x_\Lambda\|_{L^p(\mathbb{T}_\theta^d)}, \tag{7.2}$$

where $C_{\gamma,x_\Lambda} := N(supp(\widehat{x_\Lambda^\gamma}))$.

Proof. Since the case $p = q$ is clear, we consider the case $p < q$. We spilt the proof into several steps.

Step 1. We assume that $p = 2$ and $q = \infty$. By (2.6) we have

$$\|x_\Lambda\|^2_{L^2(\mathbb{T}^d_\theta)} = \tau_\theta\left(|x_\Lambda|^2\right) = \tau_\theta\left((x_\Lambda)^* x_\Lambda\right) \overset{(2.6)}{=} \sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)|^2. \tag{7.3}$$

By the triangle inequality and using the classical Cauchy-Schwarz inequality for sequences and applying formulas (2.10), (7.1), and (7.3), we obtain

$$\begin{aligned}
\|x_\Lambda\|_{L^\infty(\mathbb{T}^d_\theta)} &\overset{(7.1)}{=} \|\sum_{m\in\Lambda} \hat{x}_\Lambda(m) e^\theta_m\|_{L^\infty(\mathbb{T}^d_\theta)} \\
&\leq \sum_{m\in\Lambda} \|\widehat{x}_\Lambda(m) e^\theta_m\|_{L^\infty(\mathbb{T}^d_\theta)} \\
&= \sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)| \|e^\theta_m\|_{L^\infty(\mathbb{T}^d_\theta)} \\
&\overset{(2.10)}{=} \sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)| \cdot 1 \\
&\leq C_{x_\Lambda}^{\frac{1}{2}} (\sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)|^2)^{\frac{1}{2}} \\
&\overset{(7.3)}{=} C_{x_\Lambda}^{\frac{1}{2}} \|x_\Lambda\|_{L^2(\mathbb{T}^d_\theta)}, \qquad (7.4)
\end{aligned}$$

where $C_{x_\Lambda} = N(\mathrm{supp}(\widehat{x}_\Lambda))$.

Step 2. We now suppose that $p = 2$ and $2 < q < \infty$. Then by the inverse Hausdorff-Young inequality (see (3.2)) and the Hölder inequality with respect to $r = \frac{2}{q'}$ and $\frac{1}{r'} = 1 - \frac{q'}{2}$ and (7.3), we obtain

$$\begin{aligned}
\|x_\Lambda\|_{L^q(\mathbb{T}^d_\theta)} &\overset{(3.2)}{\leq} \|\widehat{x}_\Lambda\|_{\ell^{q'}(\mathbb{Z}^d)} \\
&= \left(\sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)|^{q'}\right)^{\frac{1}{q'}} \\
&\leq C_{x_\Lambda}^{\frac{1}{q'}-\frac{1}{2}} \left(\sum_{m\in\Lambda} |\widehat{x}_\Lambda(m)|^2\right)^{\frac{1}{2}} \\
&\overset{(7.3)}{=} C_{x_\Lambda}^{\frac{1}{2}-\frac{1}{q}} \|x_\Lambda\|_{L^2(\mathbb{T}^d_\theta)}.
\end{aligned}$$

Step 3. Let $p > 2$ and $2 < p < q \leq \infty$. We assume that γ is the smallest even integer such that $\gamma \geq \frac{p}{2}$ and $x^\gamma_\Lambda \neq 0$. Then by Lemma 7.1, $|x_\Lambda|^2$ and $|x_\Lambda|^\gamma := (x^*_\Lambda x_\Lambda)^{\gamma/2}$ are again trigonometric polynomials, respectively. Then it follows from Step 1 (see (7.4)) and the

non-commutative Hölder inequality (see [53, Theorem 5.2.9]) with $\frac{1}{2}=\frac{2}{\infty}+\frac{1}{2}$ that

$$\begin{aligned}\||x_\Lambda|^\gamma\|_{L^2(\mathbb{T}^d_\theta)} &= \||x_\Lambda|^{\gamma-p/2}|x_\Lambda|^{p/2}\|_{L^2(\mathbb{T}^d_\theta)}\\ &\le \||x_\Lambda|^{\gamma-p/2}\|_{L^\infty(\mathbb{T}^d_\theta)}\||x_\Lambda|^{p/2}\|_{L^2(\mathbb{T}^d_\theta)}\\ &= \||x_\Lambda|\|^{\gamma-p/2}_{L^\infty(\mathbb{T}^d_\theta)}\||x_\Lambda|\|^{p/2}_{L^p(\mathbb{T}^d_\theta)}\\ &= \||x_\Lambda|^\gamma\|_{L^\infty(\mathbb{T}^d_\theta)}\||x_\Lambda|\|^{-p/2}_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{p/2}_{L^p(\mathbb{T}^d_\theta)}\\ &= \||x_\Lambda|^\gamma\|_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{-p/2}_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{p/2}_{L^p(\mathbb{T}^d_\theta)}\\ &\overset{(7.4)}{\le} C^{\frac{1}{2}}_{\gamma,|x_\Lambda|}\||x_\Lambda|^\gamma\|_{L^2(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{-p/2}_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{p/2}_{L^p(\mathbb{T}^d_\theta)}.\end{aligned}$$

Here and below, we use the fact that $\|x^\kappa\|_{L^\infty(\mathbb{T}^d_\theta)}=\|x\|^\kappa_{L^\infty(\mathbb{T}^d_\theta)}$ for any normal operators $x\in L^\infty(\mathbb{T}^d_\theta)$ and power $\kappa>0$, which follows from the spectral theorem. Thus, we have

$$\|x_\Lambda\|_{L^\infty(\mathbb{T}^d_\theta)} \overset{(7.4)}{\le} C^{\frac{1}{p}}_{\gamma,x_\Lambda}\|x_\Lambda\|_{L^p(\mathbb{T}^d_\theta)}. \tag{7.5}$$

Next, suppose that $p<q<\infty$. Then it follows from (7.5) that

$$\begin{aligned}\|x_\Lambda\|_{L^q(\mathbb{T}^d_\theta)} &= \||x_\Lambda|^{1-\frac{p}{q}}|x_\Lambda|^{\frac{p}{q}}\|_{L^q(\mathbb{T}^d_\theta)}\\ &\le \||x_\Lambda|^{1-\frac{p}{q}}\|_{L^\infty(\mathbb{T}^d_\theta)}\||x_\Lambda|^{\frac{p}{q}}\|_{L^q(\mathbb{T}^d_\theta)}\\ &= \|x_\Lambda\|^{1-\frac{p}{q}}_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{\frac{p}{q}}_{L^p(\mathbb{T}^d_\theta)}\\ &\overset{(7.5)}{\le} C^{\frac{1}{p}-\frac{1}{q}}_{\gamma,x_\Lambda}\|x_\Lambda\|_{L^p(\mathbb{T}^d_\theta)},\end{aligned}$$

from which the assertion follows.

Step 4. We assume that $0<p\le 2$ and $p<q\le\infty$, and assume $x_\Lambda\neq 0$. Since $\gamma=1$, repeating the argument in the previous step we obtain

$$\|x_\Lambda\|_{L^\infty(\mathbb{T}^d_\theta)} \overset{(7.5)}{\le} C^{\frac{1}{p}}_{x_\Lambda}\|x_\Lambda\|_{L^p(\mathbb{T}^d_\theta)}. \tag{7.6}$$

Therefore, it follows from (7.6) that

$$\begin{aligned}\|x_\Lambda\|_{L^q(\mathbb{T}^d_\theta)} &= \||x_\Lambda|^{1-\frac{p}{q}}|x_\Lambda|^{\frac{p}{q}}\|_{L^q(\mathbb{T}^d_\theta)}\\ &\le \||x_\Lambda|^{1-\frac{p}{q}}\|_{L^\infty(\mathbb{T}^d_\theta)}\||x_\Lambda|^{\frac{p}{q}}\|_{L^q(\mathbb{T}^d_\theta)}\\ &= \|x_\Lambda\|^{1-\frac{p}{q}}_{L^\infty(\mathbb{T}^d_\theta)}\|x_\Lambda\|^{\frac{p}{q}}_{L^p(\mathbb{T}^d_\theta)}\\ &\overset{(7.6)}{\le} C^{\frac{1}{p}-\frac{1}{q}}_{x_\Lambda}\|x_\Lambda\|_{L^p(\mathbb{T}^d_\theta)}.\end{aligned}$$

This completes the proof. □

Remark 7.3. We are not interested in the best possible constants in (7.2) for all polynomials as x_Λ. However, if $\theta=0$, then we have the sharpness of the constant in (7.2) in the case $p=2$ and $q=\infty$, which is attained on the Dirichlet kernel (see, [39, Theorem 3.1]).

8. Besov spaces on quantum tori

In this section, we study Besov spaces from the different point of view, and analyse embedding properties of Besov spaces on $\mathbb{T}^d_\theta$ by using the Nikolskii inequality. In the definition of Besov spaces we use characteristic functions of dyadic corridors or dyadic cubes as in [49, Section 3.5.3]. Similar investigations were done in [39] for compact homogeneous manifolds. To this end we introduce

$$\begin{aligned} \Lambda_0 &:= \{n \in \mathbb{Z}^d : |n| < 2\}, \\ \Lambda_k &:= \{n \in \mathbb{Z}^d : 2^{k-1} < |n| < 2^{k+1}\}; \quad k = 1, 2, \dots, \end{aligned} \tag{8.1}$$

where $|n|$ is the Euclidean length on $\mathbb{Z}^d$. Let T^{Λ_k} be the set of all trigonometric polynomials as in (7.1). Each element $x_{\Lambda_k} \in T^{\Lambda_k}$ will be denoted briefly by x_k, that is,

$$x_0 = \widehat{x}_0(0), \;\; x_k = \sum_{2^{k-1}<|n|<2^{k+1}} \widehat{x}_k(n) e^\theta_n, \;\; n \in \mathbb{Z}^d \setminus \{0\}, \quad k = 1, 2, \dots.$$

There is also a canonical Fréchet topology on $C^\infty(\mathbb{T}^d_\theta)$. The space $\mathcal{D}'(\mathbb{T}^d_\theta)$ is defined to be the topological dual of $C^\infty(\mathbb{T}^d_\theta)$ and is called the space of distributions on $\mathbb{T}^d_\theta$. In other words, the dual space $\mathcal{D}'(\mathbb{T}^d_\theta)$ is endowed with the weak*-topology. We denote by $\langle \cdot, \cdot \rangle$ the duality bracket between $C^\infty(\mathbb{T}^d_\theta)$ and $\mathcal{D}'(\mathbb{T}^d_\theta)$, that is $F(x) := \langle F, x \rangle$ for $x \in C^\infty(\mathbb{T}^d_\theta)$ and $F \in \mathcal{D}'(\mathbb{T}^d_\theta)$. Moreover, for $1 \le p \le \infty$, the space $L^p(\mathbb{T}^d_\theta)$ is naturally embedded into $\mathcal{D}'(\mathbb{T}^d_\theta)$. Since $T^{\Lambda_k} \subset L^p(\mathbb{T}^d_\theta) \subset \mathcal{D}'(\mathbb{T}^d_\theta)$ for $1 \le p \le \infty$, it follows that all trigonometric polynomials as in (7.1) are elements of $\mathcal{D}'(\mathbb{T}^d_\theta)$. The Fourier transform extends to $\mathcal{D}'(\mathbb{T}^d_\theta)$. For $F \in \mathcal{D}'(\mathbb{T}^d_\theta)$ and $n \in \mathbb{Z}^d$, define Fourier coefficient $\widehat{F}(n) := \langle F, (e^\theta_n)^* \rangle$. Then the Fourier series of F converges to F according to any (reasonable) summation method: $F = \sum\limits_{n \in \mathbb{Z}^d} \widehat{F}(n) e^\theta_n$. For more details, see [54]. In order not to confuse the notations, we will denote the elements of $\mathcal{D}'(\mathbb{T}^d_\theta)$ again by x. The following definition is a non-commutative analogue of the definition in [49, Theorem 2, p.164].

Definition 8.1. Let $0 < p, q \le \infty$ and $r \in \mathbb{R}$. Let $\{\Lambda_k\}_{k=0}^\infty$ be the sequence of finite subsets of $\mathbb{Z}^d$ defined in (8.1). Then the associated Besov space on $\mathbb{T}^d_\theta$ is defined by

$$\begin{aligned} B^r_{p,q}(\mathbb{T}^d_\theta) &= \{x \in \mathcal{D}'(\mathbb{T}^d_\theta) : \exists \{x_k\}_{k=0}^\infty \subset C^\infty(\mathbb{T}^d_\theta) \text{ with } x_k \in T^{\Lambda_k}, k \in \mathbb{Z}_+, \text{ such that} \\ x &= \sum_{k=0}^\infty x_k \text{ in } \mathcal{D}'(\mathbb{T}^d_\theta) \text{ and } \|x\|_{B^r_{p,q}(\mathbb{T}^d_\theta)} < \infty\}, \end{aligned}$$

where

$$\|x\|_{B^r_{p,q}(\mathbb{T}^d_\theta)} := \inf \|\{x_k\}_{k\ge 0}\|_{\ell^q_r(L^p)} = \inf \left(\sum_{k=0}^\infty 2^{krq} \|x_k\|_p^q \right)^{\frac{1}{q}}, \quad q < \infty, \tag{8.2}$$

and

$$\|x\|_{B^r_{p,\infty}(\mathbb{T}^d_\theta)} := \inf \|\{x_k\}_{k\ge 0}\|_{\ell^\infty_r(L^p)} = \inf \left(\sup_{k\ge 0} 2^{kr} \|x_k\|_p \right), \quad q = \infty, \tag{8.3}$$

where the infimum is taken over all admissible representations of $\{x_k\}_{k=0}^\infty$ as above.

Remark 8.2. If $\theta = 0$, then this kind of definition was studied in [49]. Furthermore,

$$\inf \left(\sum_{k=0}^{\infty} 2^{krq} \|x_k\|_p^q \right)^{\frac{1}{q}},$$

where the infimum is taken over all admissible representations as in Definition 8.1, is an equivalent quasi-norm in Besov spaces on ordinary d-torus in terms of Littlewood-Paley decomposition (see [49, Theorem 2, p. 162]). Also, in the general case when $\theta \neq 0$ and $1 \leq p, q \leq \infty$, we conjecture that our definition is equivalent to [54, Definition 3.1]. Indeed, let φ be a Schwartz function on $\mathbb{R}^d$ such that the condition (3.1) in [54] holds. Then the sequence $\{\varphi(2^{-k}\cdot)\}_{k\geq 0}$ is a Littlewood-Paley decomposition of $\mathbb{T}^d$. As in [54, Section 3.1], the Fourier multiplier on $\mathcal{D}'(\mathbb{T}^d_\theta)$ with the symbol $\varphi(2^{-k}\cdot)$ is defined by

$$\widetilde{\varphi}_k * x = \sum_{m\in\mathbb{Z}^d} \varphi(2^{-k}m)\widehat{x}(m)e^{\theta}_m.$$

This sum is finite for every $k = 0, 1, 2, ..., .$ In particular, we have $\widetilde{\varphi}_k * x \in T^{\Lambda_k}$ whenever $\mathrm{supp}(\varphi) \subset \Lambda_k$, where Λ_k is defined by (8.1). In this case, we guess that the norm in (8.2) is equivalent to that in [54, Definition 3.1] for $1 \leq p, q \leq \infty$.

The following result describe basic properties of Besov spaces in Definition 8.1. Similar results to some of results below can be found from [54] (see for example [54, Proposition 3.3 and Theorem 6.2]) in Besov spaces on the non-commutative torus which are defined in terms of Littlewood-Paley decomposition in the case when $1 \leq p, q \leq \infty$. Our approach allows us to extend the range of indices including the case $0 < p, q < 1$.

Theorem 8.3. *Let $0 < p, q \leq \infty$ and $r \in \mathbb{R}$. We have*

(i) $B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta) \hookrightarrow B^r_{p,q_1}(\mathbb{T}^d_\theta) \hookrightarrow B^r_{p,q_2}(\mathbb{T}^d_\theta) \hookrightarrow B^r_{p,\infty}(\mathbb{T}^d_\theta)$, $\varepsilon > 0$, $0 < p \leq \infty$, $0 < q_1 \leq q_2 \leq \infty$;
(ii) $B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta) \hookrightarrow B^r_{p,q_2}(\mathbb{T}^d_\theta)$, $\varepsilon > 0$, $0 < p \leq \infty$, $1 \leq q_2 < q_1 \leq \infty$;
(iii) $B^{r_1}_{p_1,q}(\mathbb{T}^d_\theta) \hookrightarrow B^{r_2}_{p_2,q}(\mathbb{T}^d_\theta)$, $0 < p_1 \leq p_2 \leq \infty$, $0 < q < \infty$, $r_2 = r_1 - d\Big(\frac{1}{p_1} - \frac{1}{p_2}\Big)$;
(iv) $B^{r}_{p_1,1}(\mathbb{T}^d_\theta) \hookrightarrow L^{p_2}(\mathbb{T}^d_\theta)$, $0 < p_1 < p_2 \leq \infty$, $r = d\Big(\frac{1}{p_1} - \frac{1}{p_2}\Big)$;
(v) $B^{r}_{p,q}(\mathbb{T}^d_\theta) \hookrightarrow L^{q}(\mathbb{T}^d_\theta)$, $1 < p < q < \infty$, $r = d\Big(\frac{1}{p} - \frac{1}{q}\Big)$;
(vi) $B^{r}_{p,q}(\mathbb{T}^d_\theta) \hookrightarrow L^{p}(\mathbb{T}^d_\theta)$, $1 \leq p \leq \infty$, $0 < q \leq \infty$, $r > 0$;
(vii) $B^{r}_{p,q}(\mathbb{T}^d_\theta) \hookrightarrow L^{1}(\mathbb{T}^d_\theta)$, $0 < p < 1$, $0 < q \leq \infty$, $r > d(\frac{1}{p} - 1)$.

Proof. Let us prove (i). For any sequence $\{b(k)\}_{k\geq 0}$ such that $b(k) \geq 0$, $k \geq 0$ and for any $\varepsilon > 0$, we have the following simple inequality

$$\sup_{k\geq 0} b(k) \leq \Big(\sum_{k\geq 0} b(k)^{q_2} \Big)^{1/q_2} \leq \Big(\sum_{k\geq 0} b(k)^{q_1} \Big)^{1/q_1} \leq \Big(\sum_{k\geq 0} 2^{k\varepsilon q_1} b(k)^{q_1} \Big)^{1/q_1}.$$

Fix x_k and setting $b(k) := 2^{kr}\|x_k\|_p$ for $k \geq 0$ and $r \in \mathbb{R}$, and applying above inequalities we obtain

$$\begin{aligned}
\|x\|_{B^r_{p,\infty}(\mathbb{T}^d_\theta)} &\overset{(8.3)}{=} \inf\Big(\sup_{k\geq 0} 2^{kr}\big\|x_k\big\|_p\Big) \\
&\leq \inf\Big(\sum_{k\geq 0} 2^{krq_2}\|x_k\|_p^{q_2}\Big)^{1/q_2} \overset{(8.2)}{=} \|x\|_{B^r_{p,q_2}(\mathbb{T}^d_\theta)} \\
&\leq \inf\Big(\sum_{k\geq 0} 2^{krq_1}\|x_k\|_p^{q_1}\Big)^{1/q_1} \overset{(8.2)}{=} \|x\|_{B^r_{p,q_1}(\mathbb{T}^d_\theta)} \\
&\leq \inf\Big(\sum_{k\geq 0} 2^{k\varepsilon q_1}2^{krq_1}\|x_k\|_p^{q_1}\Big)^{1/q_1} \overset{(8.2)}{=} \|x\|_{B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta)}
\end{aligned}$$

for $x \in B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta)$, which is the desired result.

(ii) Let $1 \leq q_2 < q_1 < \infty$ and $\varepsilon > 0$. Then by the Hölder inequality with respect to $1 = \frac{q_2}{q_1} + \frac{q_1-q_2}{q_1} = \frac{1}{\frac{q_1}{q_2}} + \frac{1}{\frac{q_1}{q_1-q_2}}$, we have

$$\begin{aligned}
\Big(\sum_{k\geq 0} 2^{kq_2r}\|x_k\|_p^{q_2}\Big)^{1/q_2} &= \Big(\sum_{k\geq 0} \frac{2^{kq_2(r+\varepsilon)}}{2^{kq_2\varepsilon}}\|x_k\|_p^{q_2}\Big)^{1/q_2} \\
&\leq \Big(\sum_{k\geq 0} 2^{k(r+\varepsilon)q_1}\|x_k\|_p^{q_1}\Big)^{1/q_1} \cdot \Big(\sum_{k\geq 0} 2^{-\frac{kq_1q_2\varepsilon}{q_1-q_2}}\Big)^{\frac{q_1-q_2}{q_1q_2}}. \qquad (8.4)
\end{aligned}$$

Since the series $\sum_{k\geq 0} 2^{-\frac{kq_1q_2\varepsilon}{q_1-q_2}}$ converges, we have

$$\Big(\sum_{k\geq 0} 2^{-\frac{kq_1q_2\varepsilon}{q_1-q_2}}\Big)^{\frac{q_1-q_2}{q_1q_2}} \leq C < \infty$$

for some constant $C > 0$. By taking infimum from both side of (8.4) over such $\{x_k\}_{k=0}^\infty$, we find

$$\|x\|_{B^r_{p,q_2}(\mathbb{T}^d_\theta)} \leq C \cdot \|x\|_{B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta)}$$

for all $x \in B^{r+\varepsilon}_{p,q_1}(\mathbb{T}^d_\theta)$.

(iii) Let $0 < p_1 \leq p_2 \leq \infty$, $0 < q < \infty$, and $r_2 = r_1 - d\Big(\frac{1}{p_1} - \frac{1}{p_2}\Big)$. Then by the Nikolskii inequality from Theorem 7.2 we have

$$\sum_{k\geq 0} 2^{kqr_2}\|x_k\|_{p_2}^q \overset{(7.2)}{\lesssim} \sum_{k\geq 0} 2^{kqr_2}\cdot 2^{kqd(\frac{1}{p_1}-\frac{1}{p_2})}\|x_k\|_{p_1}^q = \sum_{k\geq 0} 2^{kq\left(r_2+d(\frac{1}{p_1}-\frac{1}{p_2})\right)}\|x_k\|_{p_1}^q.$$

Thus, taking infimum again over such $\{x_k\}_{k=0}^\infty$, we obtain

$$\|x\|^q_{B^{r_2}_{p_2,q}(\mathbb{T}^d_\theta)} \lesssim \|x\|^q_{B^{r_1}_{p_1,q}(\mathbb{T}^d_\theta)} \quad \forall x \in B^{r_1}_{p_1,q},(\mathbb{T}^d_\theta).$$

(iv) By the Nikolskii inequality from Theorem 7.2 we obtain

$$\|x\|_{L^{p_2}(\mathbb{T}^d_\theta)} \leq \Big\|\sum_{k\geq 0} x_k\Big\|_{p_2} \lesssim \sum_{k\geq 0}\|x_k\|_{p_2} \lesssim \sum_{k\geq 0} 2^{kd(\frac{1}{p_1}-\frac{1}{p_2})}\|x_k\|_{p_1}$$

where $r=d(\frac{1}{p_1}-\frac{1}{p_2})$. Then, taking infimum over $\{x_k\}_{k=0}^{\infty}$, we have

$$\|x\|_{L^{p_2}(\mathbb{T}^d_\theta)}\lesssim \|x\|_{B^r_{p_1,1}(\mathbb{T}^d_\theta)},\quad \forall x\in B^r_{p_1,1}(\mathbb{T}^d_\theta).$$

(v) It follows from (iv). Indeed, let T be such that $T: B^r_{p,1}(\mathbb{T}^d_\theta)\hookrightarrow L^q(\mathbb{T}^d_\theta)$. Hence, for parameters p,q, and r one can find couples (q_0,r_0), (q_1,r_1) and $\eta\in(0,1)$ satisfying the following conditions

$$d\Big(\frac{1}{p}-\frac{1}{q_0}\Big)=r_0,\quad d\Big(\frac{1}{p}-\frac{1}{q_1}\Big)=r_1,\quad r_0<r<r_1,$$

and

$$r=(1-\eta)r_0+\eta r_1,\quad \frac{1}{q}=\frac{1-\eta}{q_0}+\frac{\eta}{q_1}.$$

By (iv) we have $T: B^{r_0}_{p,1}(\mathbb{T}^d_\theta)\hookrightarrow L^{q_0}(\mathbb{T}^d_\theta)$ and $T: B^{r_1}_{p,1}(\mathbb{T}^d_\theta)\hookrightarrow L^{q_1}(\mathbb{T}^d_\theta)$. Then repeating the argument in [8, Theorem 5.6.1, p. 122], which is also true in our setting, we obtain

$$B^r_{p,q}(\mathbb{T}^d_\theta)=\Big(B^{r_0}_{p,1}(\mathbb{T}^d_\theta),B^{r_1}_{p,1}(\mathbb{T}^d_\theta)\Big)_{\eta,q}.$$

On the other hand, by [45, Formula (2.2)] we have

$$L^q(\mathbb{T}^d_\theta)=\Big(L^{q_0}(\mathbb{T}^d_\theta),L^{q_1}(\mathbb{T}^d_\theta)\Big)_{\eta,q}.$$

Thus, it follows that

$$T: B^r_{p,q}(\mathbb{T}^d_\theta)=\Big(B^{r_0}_{p,1}(\mathbb{T}^d_\theta),B^{r_1}_{p,1}(\mathbb{T}^d_\theta)\Big)_{\eta,q}\hookrightarrow\Big(L^{q_0}(\mathbb{T}^d_\theta),L^{q_1}(\mathbb{T}^d_\theta)\Big)_{\eta,q}=L^q(\mathbb{T}^d_\theta),$$

$r=d(\frac{1}{p}-\frac{1}{q})$. In other words, we have

$$B^r_{p,q}(\mathbb{T}^d_\theta)\hookrightarrow L^q(\mathbb{T}^d_\theta).$$

(vi) Let $r>0$, $1\le p\le\infty$, and $0<q\le\infty$. Then by the Hölder inequality with respect to $1=\frac{1}{q}+\frac{1}{q'}$ and (8.2), we find

$$\begin{aligned}
\|x\|_{L^p(\mathbb{T}^d_\theta)} &\le \|\sum_{k\ge0}x_k\|_p\lesssim\sum_{k\ge0}\|x_k\|_p=\sum_{k\ge0}2^{rk}2^{-rk}\|x_k\|_p\\
&\le \Big(\sum_{k\ge0}2^{-q'rk}\Big)^{\frac{1}{q'}}\Big(\sum_{k\ge0}2^{qrk}\|x_k\|_p^q\Big)^{\frac{1}{q}}\quad\text{for}\quad 1\le q<\infty\\
&\lesssim \Big(\sum_{k\ge0}2^{-rk}\Big)\sup_{k\ge0}2^{rk}\|x_k\|_p\quad\text{for}\quad q=\infty\\
&\lesssim \sum_{k\ge0}\big(2^{qrk}\|x_k\|_p^q\big)^{\frac{1}{q}}\le\big(\sum_{k\ge0}2^{qrk}\|x_k\|_p^q\big)^{\frac{1}{q}}\quad\text{for}\quad 0<q<1.
\end{aligned}$$

Here, for the case $0<q<1$, we use the inequality in [20, Exercise 1.1.4 (b), p. 11]. Then, taking infimum from both side of above inequalities over all such $\{x_k\}_{k=0}^{\infty}$, we obtain

$$\|x\|_{L^p(\mathbb{T}^d_\theta)}\lesssim\|x\|_{B^r_{p,q}(\mathbb{T}^d_\theta)},\quad x\in B^r_{p,q}(\mathbb{T}^d_\theta).$$

(vii) Let $r > d(\frac{1}{p}-1)$, $0<p<1$, and $0<q\le\infty$. Then applying the Nikolskii inequality from Theorem 7.2 with respect to Λ_k, we have

$$\|x\|_{L^1(\mathbb{T}^d_\theta)} \lesssim \sum_{k\ge 0}\|x_k\|_1 \le 2^{d(\frac{1}{p}-1)k}\sum_{k\ge 0}\|x_k\|_p \le \sum_{k\ge 0}2^{rk}\|x_k\|_p.$$

Now, repeating the argument in (vi) we obtain

$$\inf\left(\sum_{k\ge 0}2^{rk}\|x_k\|_p\right) \overset{(8.2)}{\lesssim} \|x\|_{B^r_{p,q}(\mathbb{T}^d_\theta)}.$$

Hence,

$$\|x\|_{L^1(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^r_{p,q}(\mathbb{T}^d_\theta)}, \quad x\in B^r_{p,q}(\mathbb{T}^d_\theta),$$

completing the proof. □

9. Wiener and β-Wiener spaces on quantum tori

In this section, we define quantum analogue of Wiener and β-Wiener spaces. We also study embedding properties between Besov and these spaces. For the classical case when $d=1$ and for details about these spaces we refer to Kahane's book [23] and for these spaces on homogeneous manifolds we refer the reader to [39].

Definition 9.1. Define the Wiener space on non-commutative torus as follows:

$$\begin{aligned} A(\mathbb{T}^d_\theta) \quad = \quad & \{x\in\mathcal{D}'(\mathbb{T}^d_\theta) : \exists\{x_k\}_{k=0}^\infty \subset C^\infty(\mathbb{T}^d_\theta) \text{ with } x_k\in T^{\Lambda_k}, k\in\mathbb{Z}_+ \text{ such that} \\ x \overset{\mathcal{D}'}{=} & \sum_{k=0}^\infty x_k \text{ and } \|x\|_{A(\mathbb{T}^d_\theta)} = \inf\left(\sum_{k=0}^\infty\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|\right)<\infty\}, \end{aligned}$$

where the infimum is taken over all admissible representations of such $\{x_k\}_{k=0}^\infty$.

Theorem 9.2. *We have*

$$\|x\|_{A(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^{d/2}_{2,1}(\mathbb{T}^d_\theta)}, \quad x\in B^{d/2}_{2,1}(\mathbb{T}^d_\theta).$$

Proof. Write

$$\|x\|_{A(\mathbb{T}^d_\theta)} \le \sum_{k=0}^\infty F_k,$$

where $F_k := \sum\limits_{n\in\Lambda_k}|\widehat{x}_k(n)|, k\ge 0$. Hence, by the Hölder inequality and the Plancherel identity (2.9) we have

$$F_k \le \Big(\sum_{n\in\Lambda_k}1\Big)^{1/2}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^2\Big)^{1/2} \overset{(2.9)}{\lesssim} 2^{k\frac{d}{2}}\|x_k\|_2.$$

Therefore,

$$\|x\|_{A(\mathbb{T}^d_\theta)} = \inf\left(\sum_{k=0}^\infty F_k\right) \lesssim \inf\left(\sum_{k=0}^\infty 2^{k\frac{d}{2}}\|x_k\|_2\right) := \|x\|_{B^{d/2}_{2,1}(\mathbb{T}^d_\theta)},$$

which completes the proof. □

For $0<\beta<\infty$, let us now study the β-absolute convergence of the Fourier series on non-commutative torus, which would be defined as in the case $\beta=1$ by

$$A^{\beta}(\mathbb{T}^d_\theta)=\{x\in\mathcal{D}'(\mathbb{T}^d_\theta):\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)}=\inf\Big(\sum_{k=0}^{\infty}\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\Big)^{1/\beta}<\infty\},$$

where the infimum is taken over all admissible representations of such $\{x_k\}_{k=0}^{\infty}$.

Theorem 9.3. *Let $1<p\leq 2$. Then*

$$\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)},\quad x\in B^{\alpha d}_{p,\beta}(\mathbb{T}^d_\theta),$$

for any $s>0$ and $\beta=(s+\frac{1}{p'})^{-1}$.

Proof. As previous case, write

$$\|x\|^{\beta}_{A^{\beta}(\mathbb{T}^d_\theta)}=\inf\left(\sum_{k=0}^{\infty}F_{k,\beta}\right),$$

where $F_{k,\beta}:=\sum\limits_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}, k\geq 0$. First, assume that $\beta\equiv(s+\frac{1}{p'})^{-1}\geq 2$. Applying the Hausdorff-Young inequality (3.2), we obtain

$$F_{k,\beta}=\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\overset{(3.2)}{\leq}\|x_k\|^{\beta}_{\beta'}.$$

Applying Nikolskii's inequality from Theorem 7.2 for $(L^{\beta'}(\mathbb{T}^d_\theta),L^{p}(\mathbb{T}^d_\theta))$ with respect to $\beta<p'$, or equivalently, $p<\beta'$, we obtain

$$F_{k,\beta}\leq\|x_k\|^{\beta}_{\beta'}\leq 2^{kd\beta(\frac{1}{p}-\frac{1}{\beta'})}\|x_k\|^{\beta}_{p}=2^{kd\beta(\frac{1}{\beta}-\frac{1}{p'})}\|x_k\|^{\beta}_{p}.$$

Therefore,

$$\|x\|^{\beta}_{A^{\beta}(\mathbb{T}^d_\theta)}=\inf\left(\sum_{k=0}^{\infty}F_{k,\beta}\right)\leq\inf\left(\sum_{k=0}^{\infty}2^{kd\beta(\frac{1}{\beta}-\frac{1}{p'})}\|x_k\|^{\beta}_{p}\right):=\|x\|^{\beta}_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)},$$

where $s=(\frac{1}{\beta}-\frac{1}{p'})$.

Now, let $\beta\equiv(s+\frac{1}{p'})^{-1}<2$. Set $\kappa:=\frac{p'}{p'-\beta}>1$ and $\delta:=\frac{\beta}{p'}-\frac{\beta}{2}$. Then we have

$$\frac{1}{\kappa}+\delta=1-\frac{\beta}{2}=\beta(\frac{1}{\beta}-\frac{1}{2}).$$

Hence, by the Hölder inequality we have

$$\begin{aligned}F_{k,\beta}&=\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\leq\Big(\sum_{n\in\Lambda_k}1^{\kappa}\Big)^{1/\kappa}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta\kappa'}\Big)^{1/\kappa'}\\&\lesssim 2^{k\frac{d}{\kappa}}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta\kappa'}\Big)^{1/\kappa'}=2^{kd\frac{p'-\beta}{p'}}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'}\Big)^{\beta/p'}.\end{aligned}$$

Since $p'\geq 2$, it follows from the Hausdorff-Young inequality (3.2) that

$$F_{k,\beta}\lesssim 2^{kd\frac{p'-\beta}{p'}}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'}\Big)^{\beta/p'}\overset{(3.2)}{\lesssim}2^{kd\frac{p'-\beta}{p'}}\|x_k\|^{\beta}_{p}.$$

In other words, we have

$$\begin{aligned}\|x\|^{\beta}_{A^{\beta}(\mathbb{T}^d_\theta)} &= \inf\left(\sum_{k=0}^{\infty} F_{k,\beta}\right) \leq \inf\left(\sum_{k=1}^{\infty} 2^{kd\frac{p'-\beta}{p'}}\|x_k\|_p^{\beta}\right)\\ &= \inf\left(\sum_{k=0}^{\infty} 2^{kd\beta(\frac{1}{\beta}-\frac{1}{p'})}\|x_k\|_p^{\beta}\right) := \|x\|^{\beta}_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)}\end{aligned}$$

for all $x \in B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)$, where $s = (\frac{1}{\beta} - \frac{1}{p'})$. □

Remark 9.4. (i) In the special case, when $\theta = 0$ and $d = 1$, Theorem 9.3 gives the result due to Szasz [42, p. 119].

(ii) In the case $p = 2$, we obtain the strongest result which follows from the combination of Theorem 9.3 and Theorem 8.3 (iii), that is,

$$\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^{s^*d}_{2,\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)},$$

where $s^*, s > 0$ and $\beta = (s^* + \frac{1}{2})^{-1} = (s + \frac{1}{p'})^{-1}$.

The following is the converse result to Theorem 9.3.

Theorem 9.5. *Let $2 \leq p < \infty$. Then*

$$\|x\|_{B^{\tilde{s}d}_{p,\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{A^{\beta}(\mathbb{T}^d_\theta)}, \quad x \in A^{\beta}(\mathbb{T}^d_\theta),$$

for $\tilde{s} := \min\{s, 0\}$ and $\beta = (s + \frac{1}{p'})^{-1} > 0$.

Proof. Let $\beta \leq p'$ (or equivalently $s \geq 0$). In this case $\tilde{s} = 0$. Since $p \geq 2$, applying the Hausdorff-Young inequality (3.2) we obtain

$$\begin{aligned}\|x\|^{\beta}_{B^{0}_{p,\beta}(\mathbb{T}^d_\theta)} &= \inf\left(\sum_{k=0}^{\infty}\|x_k\|_p^{\beta}\right) \overset{(3.2)}{\leq} \inf\left(\sum_{k=0}^{\infty}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'}\Big)^{\beta/p'}\right)\\ &\leq \inf\left(\sum_{k=0}^{\infty}\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\right) := \|x\|^{\beta}_{A^{\beta}(\mathbb{T}^d_\theta)}, \quad x \in A^{\beta}(\mathbb{T}^d_\theta),\end{aligned}$$

where in the third line we have used the inequality in [20, Exercise 1.1.4 (a), p. 11] for $0 < \frac{\beta}{p'} \leq 1$. Let now $\beta > p'$ (or equivalently $\beta' < p$). In this case $\tilde{s} = s < 0$. By using the Hölder inequality with respect to $\frac{\beta}{\beta - p'}$, we have

$$\begin{aligned}\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'} &\leq \Big(\sum_{n\in\Lambda_k}1^{\frac{\beta}{\beta-p'}}\Big)^{1-\frac{p'}{\beta}}\Big(\sum_{n\in\Lambda_k}(|\widehat{x}_k(n)|^{p'})^{\frac{\beta}{p'}}\Big)^{p'/\beta}\\ &\lesssim 2^{kd(1-\frac{p'}{\beta})}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\Big)^{p'/\beta}.\end{aligned}$$

Hence, by this and the Hausdorff-Young inequality (3.1) we obtain

$$\begin{aligned}
\|x\|^{\beta}_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)} &= \inf\left(\sum_{k=0}^{\infty} 2^{ksd\beta}\|x_k\|_p^{\beta}\right)\\
&\overset{(3.2)}{\leq} \inf\left(\sum_{k=0}^{\infty} 2^{ksd\beta}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'}\Big)^{\beta/p'}\right)\\
&\lesssim \inf\left(\sum_{k=0}^{\infty} 2^{ksd\beta}\Big(2^{kd(1-\frac{p'}{\beta})}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\Big)^{p'/\beta}\Big)^{\beta/p'}\right)\\
&= \inf\left(\sum_{k=0}^{\infty} 2^{ksd\beta}2^{kd(\frac{\beta}{p'}-1)}\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\right)\\
&= \inf\left(\sum_{k=0}^{\infty}\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{\beta}\right)\\
&:= \|x\|^{\beta}_{A^{\beta}(\mathbb{T}^d_\theta)}, \quad x\in A^{\beta}(\mathbb{T}^d_\theta),
\end{aligned}$$

where we used the fact $s\beta+\frac{\beta}{p'}-1=0$ in the fourth line. □

Remark 9.6. (i) If $\beta\geq 2$, then we have the strongest result when $p=2$, that is,

$$\|x\|_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{B^{s^*d}_{2,\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)},$$

where $\beta=(s^*+\frac{1}{2})^{-1}=(s+\frac{1}{p'})^{-1}$. Note that in this case $s^*,s<0$. In other words, $\tilde{s}=s$ and $\tilde{s^*}=s^*$.

(ii) If $\beta\geq 2$, then we have the strongest result when $p=\beta'$, that is,

$$\|x\|_{B^{sd}_{p,\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{B^{0}_{\beta',\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)},$$

where parameters are as in the previous case.

In both cases the result follows from the combination of Theorem 9.5 and Theorem 8.3 (iii).

10. Beurling and β-Beurling spaces on quantum tori

In this section, we define Beurling and β-Beurling spaces on quantum tori and study their embedding properties. In the classical case ($\theta=0$ and $d=1$), these spaces were introduced by Beurling [9] (see also [6] for the algebraic properties), and [39] for homogeneous manifolds.

Definition 10.1. Define the Beurling space on non-commutative torus as follows:

$$\begin{aligned}
A^*(\mathbb{T}^d_\theta) &= \{x\in\mathcal{D}'(\mathbb{T}^d_\theta):\exists\{x_k\}_{k=0}^{\infty}\subset C^{\infty}(\mathbb{T}^d_\theta) \text{ with } x_k\in T^{\Lambda_k}, k\in\mathbb{Z}_+ \text{ such that,}\\
x\overset{\mathcal{D}'}{=}&\sum_{k=0}^{\infty}x_k \text{ and } \|x\|_{A^*(\mathbb{T}^d_\theta)}=\inf\left(\sum_{k=0}^{\infty}2^{kd}\sup_{n\in\Lambda_k}|\widehat{x}_k(n)|\right)<\infty\},
\end{aligned}$$

where the infimum is taken over all admissible representations of such $\{x_k\}_{k=0}^{\infty}$.

Similar to the previous section, for any $0<\beta<\infty$ define β-version of this space as follows

$$A^{*,\beta}(\mathbb{T}^d_\theta)=\{x\in\mathcal{D}'(\mathbb{T}^d_\theta):\|x\|_{A^{*,\beta}(\mathbb{T}^d_\theta)}:=\inf\Big(|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\Big(\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\Big)^\beta\Big)^{1/\beta}<\infty\}.$$

In this case, for any $0<\beta<\infty$ we have the following expression of the norm

$$\begin{aligned}\|x\|^\beta_{A^{*,\beta}(\mathbb{T}^d_\theta)}&:=\inf\left(|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\Big(\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\Big)^\beta\right)\\ &\asymp\inf\left(|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\Big(\sup_{n\in\Lambda_k}|\widehat{x}_k(n)|\Big)^\beta\right).\end{aligned}\tag{10.1}$$

Indeed, the inequality " $\geq$ " is trivial. Let us prove the converse inequality. We write

$$\begin{aligned}|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\Big(\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\Big)^\beta&\asymp|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\sup_{k\leq l}\sup_{n\in\Lambda_l}|\widehat{x}_l(n)|^\beta\\ &\lesssim|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\sum_{l=k}^{\infty}\sup_{n\in\Lambda_l}|\widehat{x}_l(n)|^\beta\\ &=|\widehat{x}_0(0)|^\beta+\sum_{l=1}^{\infty}\sup_{n\in\Lambda_l}|\widehat{x}_l(n)|^\beta\sum_{k=1}^{l}2^{kd}\\ &\asymp|\widehat{x}_0(0)|^\beta+\sum_{l=1}^{\infty}2^{ld}\sup_{n\in\Lambda_l}|\widehat{x}_l(n)|^\beta,\end{aligned}$$

which is the desired result. So, we can work with either of equivalent expressions in (10.1). The following result shows embedding properties between Beurling and Besov spaces.

Theorem 10.2. *Let $0<\beta<\infty$ and $p\geq 2$. Then we have*

$$\|x\|_{B^{d(\frac{1}{\beta}-\frac{1}{p'})}_{p,\beta}(\mathbb{T}^d_\theta)}\lesssim\|x\|_{A^{*,\beta}(\mathbb{T}^d_\theta)},\quad x\in A^{*,\beta}(\mathbb{T}^d_\theta).$$

Proof. By the Hausdorff-Young inequality (3.1) we have

$$\begin{aligned}\|x\|_{B^{d(\frac{1}{\beta}-\frac{1}{p'})}_{p,\beta}(\mathbb{T}^d_\theta)}&=\inf\Big(|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd(1-\frac{\beta}{p'})}\Big\|\sum_{n\in\Lambda_k}\widehat{x}_k(n)e^\theta_n\Big\|^\beta_p\Big)^{1/\beta}\\ &\leq\inf\Big[|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd(1-\frac{\beta}{p'})}\Big(\sum_{n\in\Lambda_k}|\widehat{x}_k(n)|^{p'}\Big)^{\beta/p'}\Big]^{1/\beta}\\ &\lesssim\inf\Big[|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd(1-\frac{\beta}{p'})}\Big(\sum_{n\in\Lambda_k}1\Big)^{\beta/p'}\Big(\sup_{n\in\Lambda_k}|\widehat{x}_k(n)|\Big)^\beta\Big]^{1/\beta}\\ &\lesssim\inf\Big[|\widehat{x}_0(0)|^\beta+\sum_{k=1}^{\infty}2^{kd}\Big(\sup_{n\in\Lambda_k}|\widehat{x}_k(n)|\Big)^\beta\Big]^{1/\beta}\\ &:=\|x\|_{A^{*,\beta}(\mathbb{T}^d_\theta)},\quad x\in A^{*,\beta}(\mathbb{T}^d_\theta),\end{aligned}$$

which concludes the proof. □

In the case $p=2$, we obtain the following result which follows from Theorems 9.3 and 10.2.

Corollary 10.3. *Let $0<\beta<\infty$. Then*

$$\|x\|_{A^{\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^{d(\frac{1}{\beta}-\frac{1}{2})}_{2,\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{A^{*,\beta}(\mathbb{T}^d_\theta)}, \quad x\in A^{*,\beta}(\mathbb{T}^d_\theta).$$

In particular, if $\beta=1$, then

$$\|x\|_{A(\mathbb{T}^d_\theta)} \lesssim \|x\|_{B^{d/2}_{2,\beta}(\mathbb{T}^d_\theta)} \lesssim \|x\|_{A^{*}(\mathbb{T}^d_\theta)}, \quad x\in A^{*}(\mathbb{T}^d_\theta).$$

11. Interpolation

Let X_0, X_1 be two quasi-Banach spaces such that X_1 is continuously embedded in X_0 : $X_1\hookrightarrow X_0$. We define the K-functional for $x\in X_0+X_1$ by

$$K(x,t;X_0,X_1):=\inf_{x=z+y}\Big(\|z\|_{X_0}+t\|y\|_{X_1}\Big), \quad t\geq 0.$$

To study intermediate space X for the pair (X_0,X_1), that is, $X_0\cap X_1\subset X\subset X_0+X_1$, one uses the η, q-interpolation spaces. Let $0<\eta<1$. Then we define

$$(X_0,X_1)_{\eta,q}:=\Big\{x\in X_0+X_1 : \|x\|_{(X_0,X_1)_{\eta,q}}:=\Big(\int\limits_0^{\infty}\big(t^{-\eta}K(x,t)\big)^q\frac{dt}{t}\Big)^{\frac{1}{q}}<\infty\Big\},$$

for $0<q<\infty$;

$$(X_0,X_1)_{\eta,\infty}:=\{x\in X_0+X_1 : \|x\|_{(X_0,X_1)_{\eta,\infty}}:=\sup_{0<t<\infty}t^{-\eta}K(x,t)<\infty\},$$

for $q=\infty$.

Definition 11.1. We define the following Beurling-type spaces on quantum tori:

$$\begin{aligned} A^{*,\beta}_r(\mathbb{T}^d_\theta) &= \{x\in\mathcal{D}'(\mathbb{T}^d_\theta) : \exists\{x_k\}_{k=0}^{\infty}\subset C^{\infty}(\mathbb{T}^d_\theta) \text{ with } x_k\in T^{\Lambda_k}, k\in\mathbb{Z}_+ \text{ such that}\\ x&\overset{\mathcal{D}'}{=}\sum_{k=0}^{\infty}x_k \text{ and } \|x\|_{A^{*,\beta}_r(\mathbb{T}^d_\theta)}<\infty\},\end{aligned}$$

where

$$\|x\|_{A^{*,\beta}_r(\mathbb{T}^d_\theta)}:=\inf\Big(|\widehat{x}_0(0)|^{\beta}+\sum_{k=1}^{\infty}\big(2^{rkd}\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\big)^{\beta}\Big)^{\frac{1}{\beta}}, \tag{11.1}$$

and the infimum is taken over all admissible representations of such $\{x_k\}_{k=0}^{\infty}$.

Note that $A^{*,\beta}_{\frac{1}{\beta}}(\mathbb{T}^d_\theta)=A^{*,\beta}(\mathbb{T}^d_\theta)$, where $A^{*,\beta}(\mathbb{T}^d_\theta)$ is defined in Section 9.

Remark 11.2. It can be proved that β-Wiener, β-Beurling, and the space in Definition 11.1 are quasi-normed spaces. Since we do not use their completeness elsewhere, we do not try to prove that. But, we conjecture that these spaces are indeed quasi-Banach. Although we give the definition of interpolation spaces for general quasi-Banach spaces, in the theorem below we use only their quasi-norms.

Theorem 11.3. *Let $0<r_1<r_0<\infty$, $0<\beta_0,\beta_1,q\leq\infty$, and $r=(1-\eta)r_0+\eta r_1$.*

(i) *We have*

$$(A_r^{*,\beta_0}(\mathbb{T}_\theta^d), A_r^{*,\beta_1}(\mathbb{T}_\theta^d))_{\eta,q} = A_r^{*,q}(\mathbb{T}_\theta^d).$$

In particular,

$$(A^{*,\frac{1}{r_0}}(\mathbb{T}_\theta^d), A^{*,\frac{1}{r_1}}(\mathbb{T}_\theta^d))_{\eta,\frac{1}{r}} = A^{*,\frac{1}{r}}(\mathbb{T}_\theta^d);$$

(ii) *If* $0 < p \le \infty$. *Then*

$$(B_{p,\beta_0}^{r_0}(\mathbb{T}_\theta^d), B_{p,\beta_1}^{r_1}(\mathbb{T}_\theta^d))_{\eta,q} = B_{p,q}^{r}(\mathbb{T}_\theta^d).$$

Proof. Let us prove (i). We assume $x \in (A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d), A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d))_{\eta,q}$. Take any representation $x = z + y$ such that $z \in A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)$, and $y \in A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)$. Then for any $k \ge 1$, we get

$$\begin{aligned}
2^{rkd} \sup_{2^{k-1}<|n|} |\widehat{x}_k(n)| &\le \sup_{2^{k-1}<|n|} \big[(2^{(r-r_0)kd}(2^{r_0kd}|\widehat{z}_k(n)|) + 2^{(r-r_1)kd}(2^{r_1kd}|\widehat{y}_k(n)|)\big] \\
&\lesssim 2^{(r-r_0)kd}\big(|\widehat{z}_0(0)|^{\beta_0} + \big(2^{r_0kd} \sup_{2^{k-1}<|n|} |\widehat{z}_k(n)|\big)^{\beta_0}\big)^{\frac{1}{\beta_0}} \\
&+ 2^{(r-r_1)kd}\big(|\widehat{y}_0(0)|^{\beta_1} + \big(2^{r_1kd} \sup_{2^{k-1}<|n|} |\widehat{y}_k(n)|\big)^{\beta_1}\big)^{\frac{1}{\beta_1}} \\
&\le 2^{(r-r_0)kd}\Big[\Big(|\widehat{z}_0(0)|^{\beta_0} + \sum_{s=1}^{\infty}\big(2^{r_0sd} \sup_{2^{s-1}<|n|} |\widehat{z}_k(n)|\big)^{\beta_0}\Big)^{\frac{1}{\beta_0}} \\
&+ 2^{r'kd}\Big(|\widehat{y}_0(0)|^{\beta_1} + \sum_{s=1}^{\infty}\big(2^{r_1sd} \sup_{2^{s-1}<|n|} |\widehat{y}_k(n)|\big)^{\beta_1}\Big)^{\frac{1}{\beta_1}}\Big] \\
&\overset{(11.1)}{=} 2^{(r-r_0)kd}\Big(\|z\|_{A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)} + 2^{r'kd}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}\Big) \qquad (11.2)
\end{aligned}$$

and

$$\begin{aligned}
|\widehat{x}_0(0)| &\lesssim 2^{(r-r_0)d}\big(|\widehat{z}_0(0)|^{\beta_0} + \big(2^{r_0d} \sup_{2^{k-1}<|n|} |\widehat{z}_k(n)|\big)^{\beta_0}\big)^{\frac{1}{\beta_0}} \\
&+ 2^{(r-r_1)d}\big(|\widehat{z}_0(0)|^{\beta_1} + \big(2^{r_1d} \sup_{2^{k-1}<|n|} |\widehat{z}_k(n)|\big)^{\beta_1}\big)^{\frac{1}{\beta_1}} \\
&\lesssim 2^{(r-r_0)d}\Big(\|z\|_{A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)} + 2^{r'd}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}\Big), \qquad (11.3)
\end{aligned}$$

where $0 < r_1 < r_0 < \infty$, $0 < \beta_0, \beta_1 \le \infty$, and $r' = r_0 - r_1$, $r = (1-\eta)r_0 + \eta r_1$.

Using (11.2) and (11.3) and the definition $K(x,t) := K(x,t; A_{r_0}^{*,\beta_0}, A_{r_1}^{*,\beta_1})$, we find

$$2^{rdk} \sup_{2^{k-1}<|n|} |\widehat{x}_k(n)| \overset{(11.2)}{\lesssim} 2^{(r-r_0)kd}K(x, 2^{r'kd}) = 2^{-\eta r'kd}K(x, 2^{r'kd})$$

and

$$|\widehat{x}_0(0)| \overset{(11.3)}{\lesssim} 2^{(r-r_0)d}K(x, 2^{r'd}) = 2^{-\eta r'd}K(x, 2^{r'd}),$$

where $r'=r_1-r_0$. Therefore,

$$\begin{aligned}
\|x\|_{A^{*,q}_r(\mathbb{T}^d_\theta)} &\overset{(11.1)}{=} \inf\Big(|\widehat{x}_0(0)|^q+\sum_{k=1}^{\infty}\big(2^{rkd}\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\big)^q\Big)^{\frac{1}{q}}\\
&\lesssim \Big(\big(2^{-\eta r'd}K(x,2^{(r_0-r_1)d})\big)^q+\sum_{k=1}^{\infty}\big(2^{-\eta r'kd}K(x,2^{r'kd})\big)^q\Big)^{\frac{1}{q}}\\
&\le \Big(\sum_{k=1}^{\infty}\big(2^{-\eta r'kd}K(x,2^{r'kd})\big)^q+\sum_{k=1}^{\infty}\big(2^{-\eta r'kd}K(x,2^{r'kd})\big)^q\Big)^{\frac{1}{q}}\\
&= 2^{\frac{1}{q}}\Big(\sum_{k=1}^{\infty}\big(2^{-\eta r'kd}K(x,2^{r'kd})\big)^q\Big)^{\frac{1}{q}}\\
&\asymp \Big(\int_1^{\infty}\big(t^{-\eta r'd}K(x,t^{r'd})\big)^q\frac{dt}{t}\Big)^{\frac{1}{q}}\\
&\le \Big(\int_0^{\infty}\big(t^{-\eta r'd}K(x,t^{r'd})\big)^q\frac{dt}{t}\Big)^{\frac{1}{q}}\\
&\overset{(11.1)}{=} \|x\|_{\left(A^{*,\beta_0}_{r_0},A^{*,\beta_1}_{r_1}\right)_{\eta,q}},
\end{aligned}$$

which means that $\left(A^{*,\beta_0}_{r_0}(\mathbb{T}^d_\theta),A^{*,\beta_1}_{r_1}(\mathbb{T}^d_\theta)\right)_{\eta,q}\hookrightarrow A^{*,q}_r(\mathbb{T}^d_\theta)$.

Let us show the inverse embedding. Let $x\in A^{*,q}_r(\mathbb{T}^d_\theta)$, $\tau=\min\{\beta_0,\beta_1,q\}$, $r\in\mathbb{Z}^+$, and $r'=r_1-r_0$. For $l\in\mathbb{N}$ we denote

$$z:=\sum_{|n|\le 2^l}|\widehat{x}_k(n)|e^{\theta}_n,\quad y:=x-z.$$

Thus,

$$\begin{aligned}
\|z\|_{A^{*,\beta_0}_{r_0}(\mathbb{T}^d_\theta)} &\overset{(11.1)}{=} \inf\Big(|\widehat{z}_0(0)|^{\beta_0}+\sum_{k=1}^{\infty}\big(2^{r_0kd}\sup_{2^{k-1}<|n|}|\widehat{z}_k(n)|\big)^{\beta_0}\Big)^{\frac{1}{\beta_0}}\\
&\le \Big(|\widehat{z}_0(0)|^{\tau}+\sum_{k=1}^{\infty}\big(2^{r_0kd}\sup_{2^{k-1}<|n|}|\widehat{z}_k(n)|\big)^{\tau}\Big)^{\frac{1}{\tau}}\\
&\le \Big(|\widehat{x}_0(0)|^{\tau}+\sum_{k=1}^{l}\big(2^{r_0kd}\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\big)^{\tau}\Big)^{\frac{1}{\tau}} \qquad (11.4)
\end{aligned}$$

and

$$\begin{aligned}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)} &\lesssim 2^{r_1 ld}\sup_{2^l\le|n|}|\widehat{x}_l(n)| + \Big(|\widehat{y}_0(0)|^\tau + \sum_{k=l+1}^{\infty}\big(2^{r_1kd}\sup_{2^{k-1}\le|n|}|\widehat{x}_k(n)|\big)^\tau\Big)^{\frac{1}{\tau}}\\ &\lesssim 2^{r'ld}\Big(|\widehat{x}_0(0)|^\tau + \sum_{k=0}^{l}\big(2^{r_0kd}\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\big)^\tau\Big)^{\frac{1}{\tau}}\\ &+ \Big(|\widehat{x}_0(0)|^\tau + \sum_{k=l+1}^{\infty}\big(2^{r_1kd}\sup_{2^{k-1}<|n|}|\widehat{x}_k(n)|\big)^\tau\Big)^{\frac{1}{\tau}}. \qquad (11.5)\end{aligned}$$

Since

$$K(x,t^{r'd}) = \inf_{x=z+y}\big(\|z\|_{A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)} + t^{r'd}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}\big) \le t^{r'd}\|x\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)},$$

we have

$$\begin{aligned}\|x\|_{\left(A_{r_0}^{*,\beta_0},A_{r_1}^{*,\beta_1}\right)_{\eta,q}} &= \Big(\int_0^\infty\big(t^\eta K(x,t)\big)^q\frac{dt}{t}\Big)^{\frac{1}{q}}\\ &= \frac{1}{r'}\Big(\int_0^\infty\big(t^{-\eta r'd}K(x,t^{r'd})\big)^q\frac{dt}{t}\Big)^{\frac{1}{q}}\\ &\lesssim \Big(\|x\|^q_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}\int_0^1\big(t^{-\eta r'd}t^{r'd}\big)^q\frac{dt}{t} + \int_1^\infty(t^{-\eta r'd}K(x,t^{r'd}))^q\frac{dt}{t}\Big)^{\frac{1}{q}}\\ &\asymp \|x\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)} + \Big(\sum_{s=0}^{\infty}\big(2^{-\eta r'sd}K(x,2^{r'sd})\big)^q\Big)^{\frac{1}{q}}. \qquad (11.6)\end{aligned}$$

In view of $r_1 < r$, we obtain $\|x\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)} \lesssim \|x\|_{A_r^{*,\beta}(\mathbb{T}_\theta^d)}$. Then, by

$$K(x,2^{r'sd}) \le \|z\|_{A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)} + 2^{r'sd}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}$$

and formulas (11.4)-(11.6), we infer

$$\begin{aligned}\|x\|_{(A_{r_0}^{*,\beta_0},A_{r_1}^{*,\beta_1})_{\eta,q}} &\overset{(11.6)}{\lesssim} \|x\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)} + \inf_{x=z+y}\Big(\sum_{s=0}^{\infty}\big[2^{-\eta r'sd}\big(\|z\|_{A_{r_0}^{*,\beta_0}(\mathbb{T}_\theta^d)} + 2^{r'sd}\|y\|_{A_{r_1}^{*,\beta_1}(\mathbb{T}_\theta^d)}\big)\big]^q\Big)^{\frac{1}{q}}\\ &\overset{(11.4)(11.5)}{\lesssim} \|x\|_{A_r^{*,\beta}(\mathbb{T}_\theta^d)} + \inf_{x=z+y}\Bigg(\sum_{k=0}^{\infty}\Big(2^{-\eta r'qkd}2^{r'ld}\Big[|\widehat{x}_0(0)|^\tau\\ &+ \sum_{k=0}^{l}\big(2^{r_0kd}\sup_{2^k<|n|}|\widehat{x}_k(n)|\big)^\tau\Big]^{\frac{1}{\tau}}\\ &+ \Big[|\widehat{x}_0(0)|^\tau + \sum_{k=l+1}^{\infty}\big(2^{r_1kd}\sup_{2^k\le|n|}|\widehat{x}_k(n)|\big)^\tau\Big]^{\frac{1}{\tau}}\Big)^q\Bigg)^{\frac{1}{q}}.\end{aligned}$$

Further, since $\tau \le q$, applying Hardy's inequality (see [7, Theorem 2 (iv)]) we obtain

$$\|x\|_{(A_{r_0}^{*,\beta_0},A_{r_1}^{*,\beta_1})_{\eta,q}} \lesssim \|x\|_{A_r^{*,\beta}(\mathbb{T}_\theta^d)}.$$

This completes the proof of (i). To show (ii), we first note that

$$(B^{r_0}_{p,\beta_0}(\mathbb{T}^d_\theta), B^{r_1}_{p,\beta_1}(\mathbb{T}^d_\theta))_{\eta,q} \hookrightarrow B^r_{p,q}(\mathbb{T}^d_\theta)$$

can be proved similarly to the proof of (i) using the embedding

$$B^r_{p,\beta}(\mathbb{T}^d_\theta) \hookrightarrow B^r_{p,\infty}(\mathbb{T}^d_\theta)$$

which follows from Theorem 8.3 (ii). □

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11.2. **Data availability.** No new data were created or analysed during this study. Data sharing is not applicable to this article.

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